

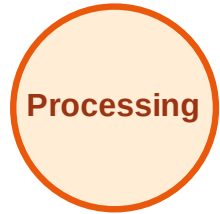
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



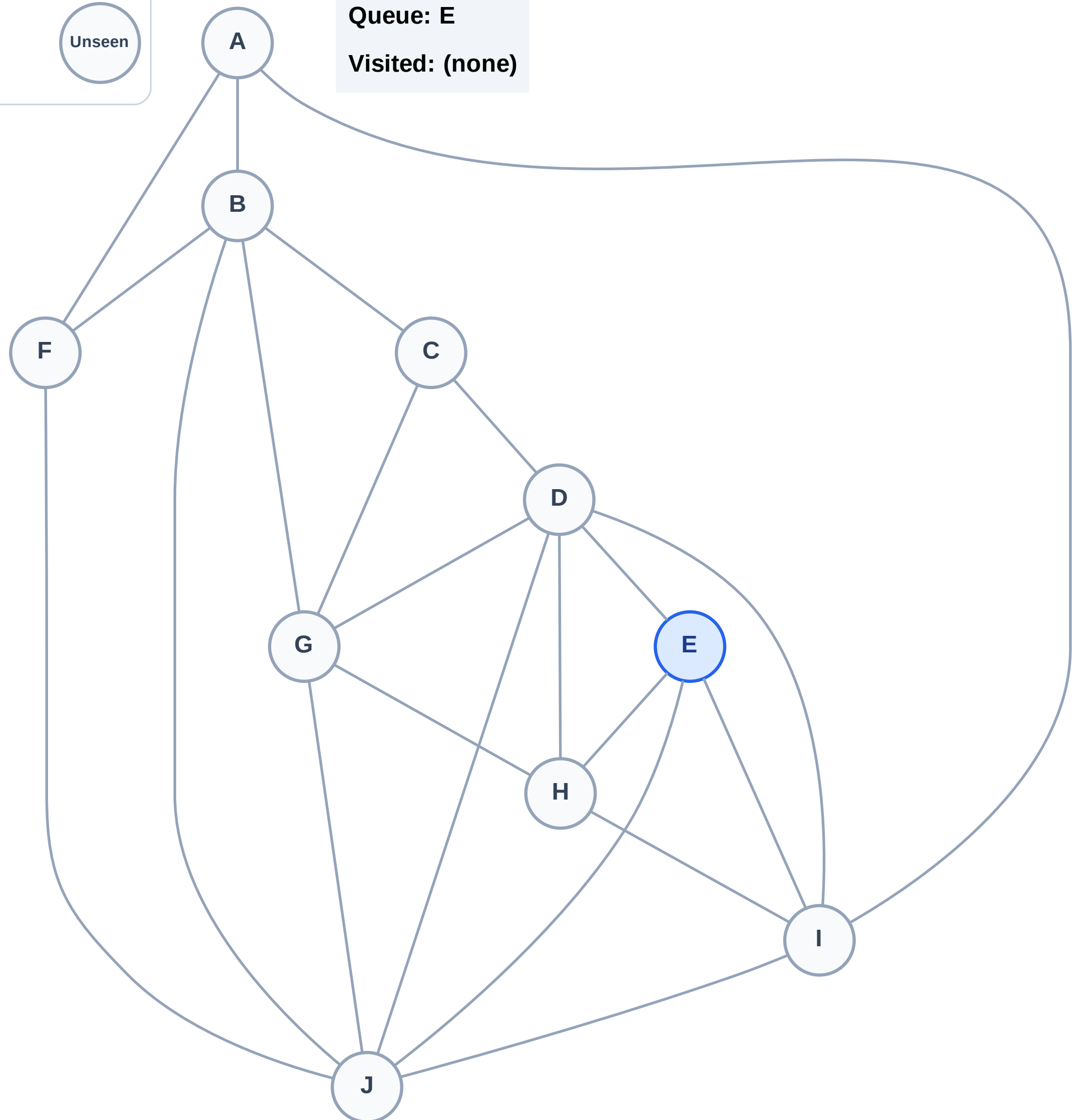
Discovered



Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

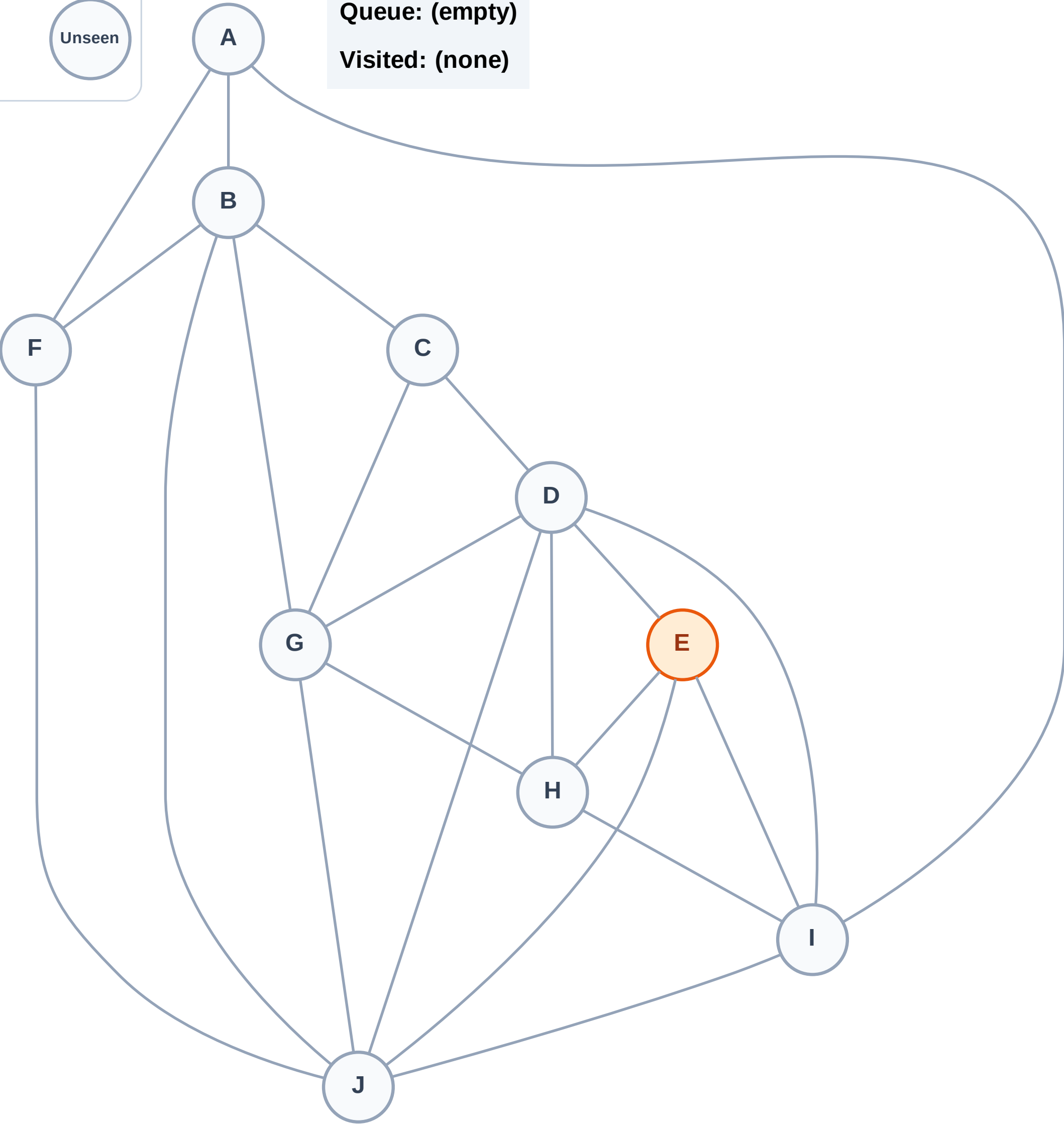
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

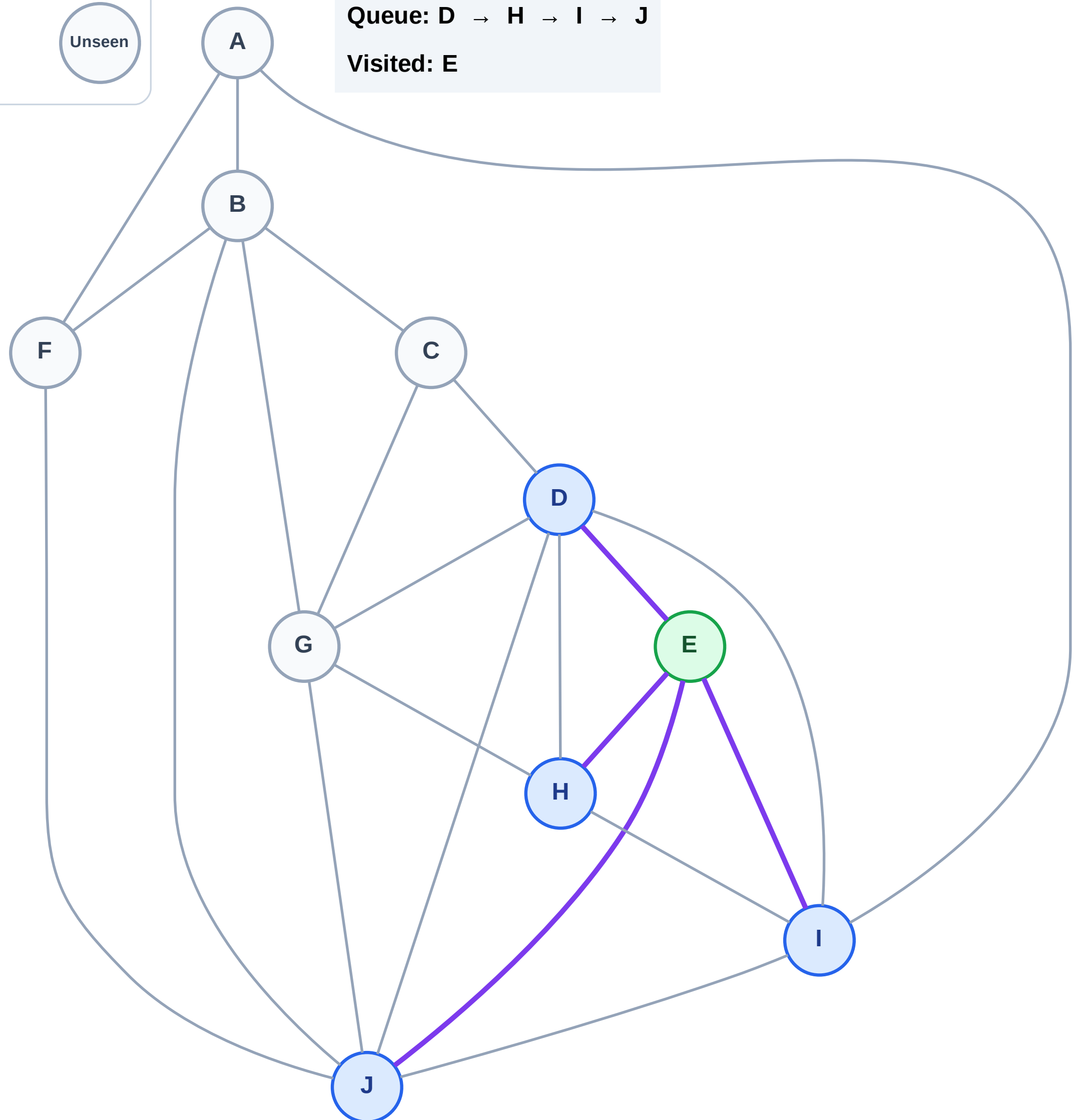
Step 3: Discover D, H, I, J from E

Legend



Queue: D → H → I → J

Visited: E



BFS Traversal

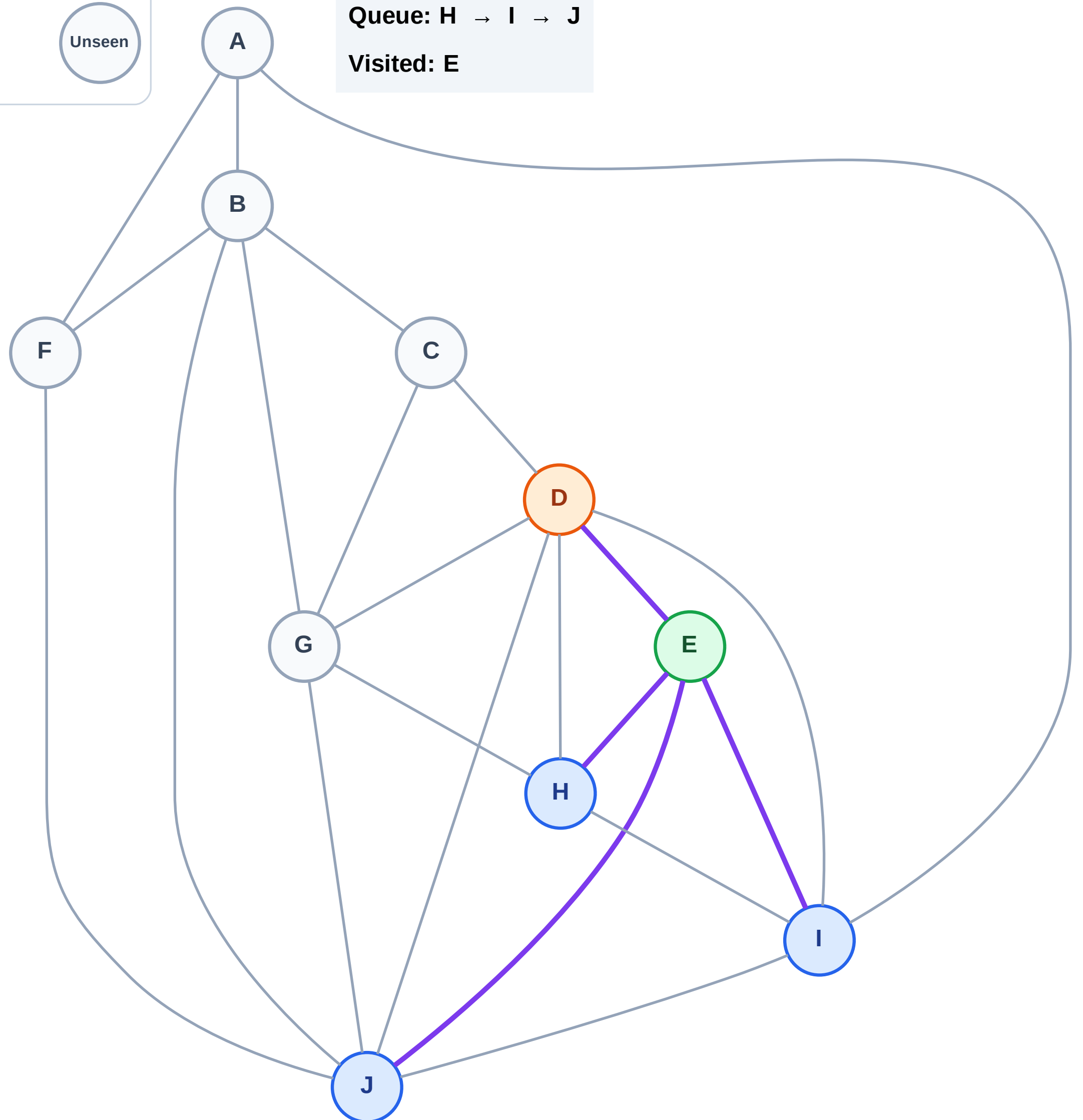
Step 4: Process node D

Legend



Queue: H → I → J

Visited: E



BFS Traversal

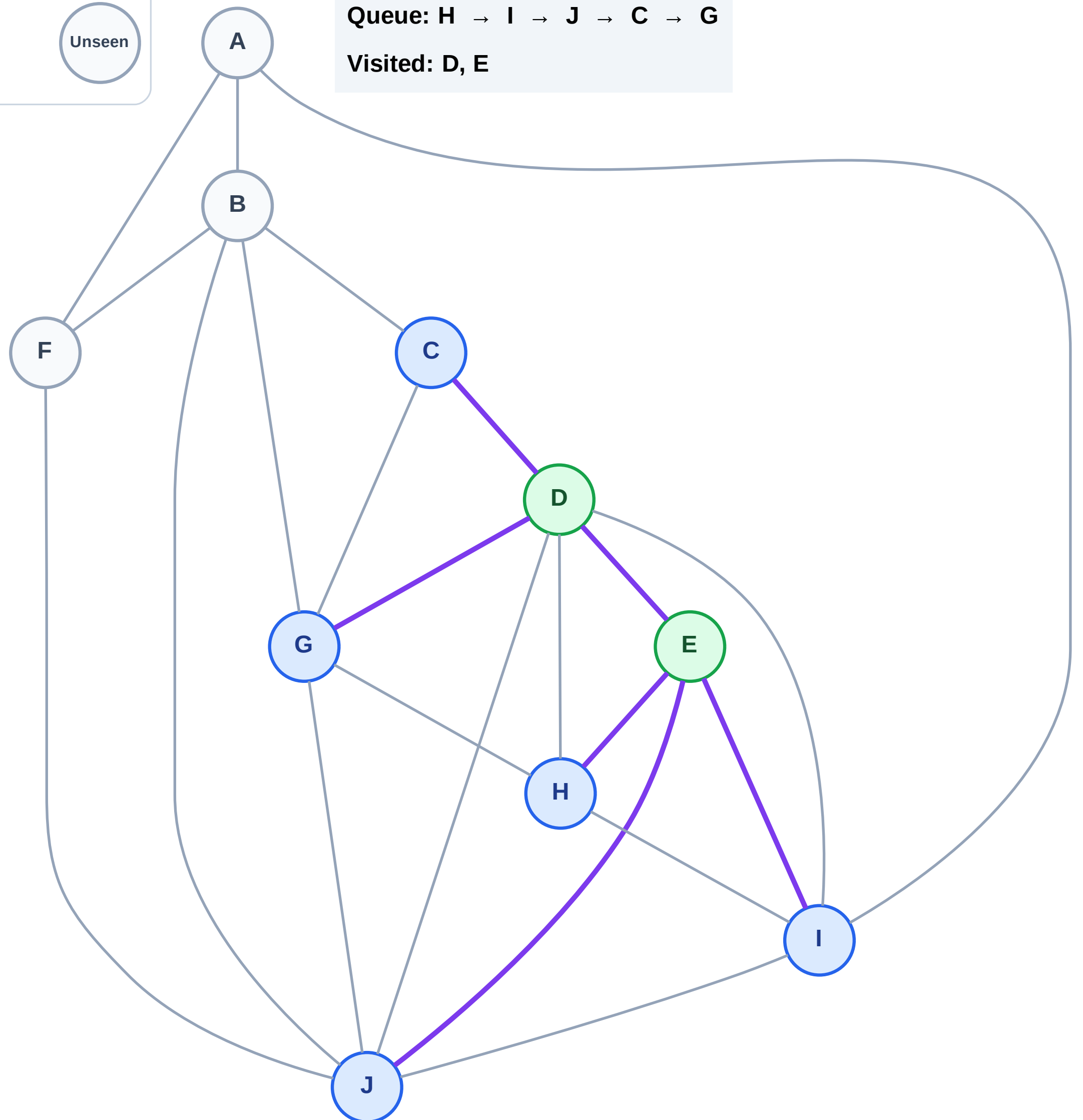
Step 5: Discover C, G from D

Legend



Queue: H → I → J → C → G

Visited: D, E

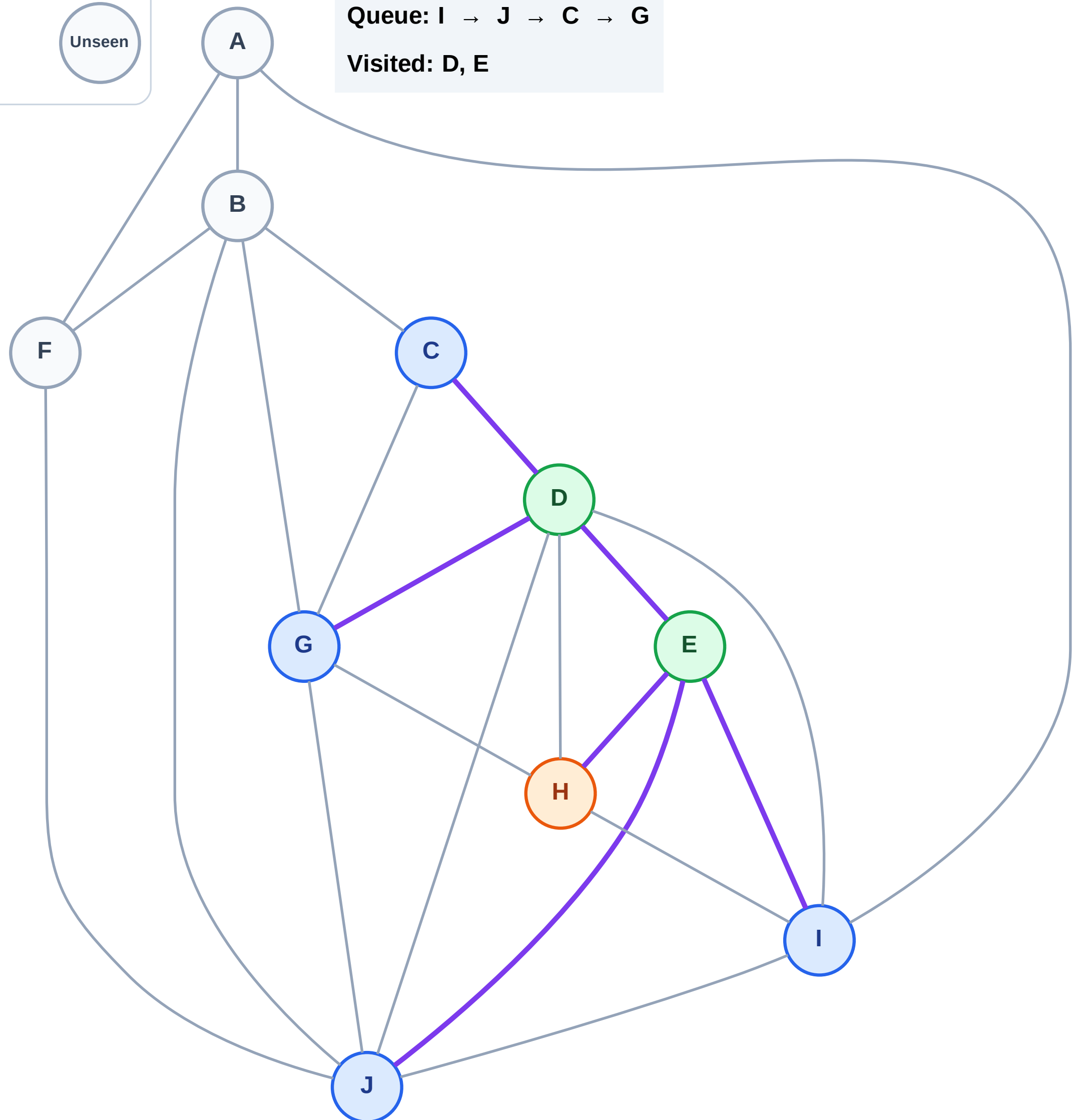
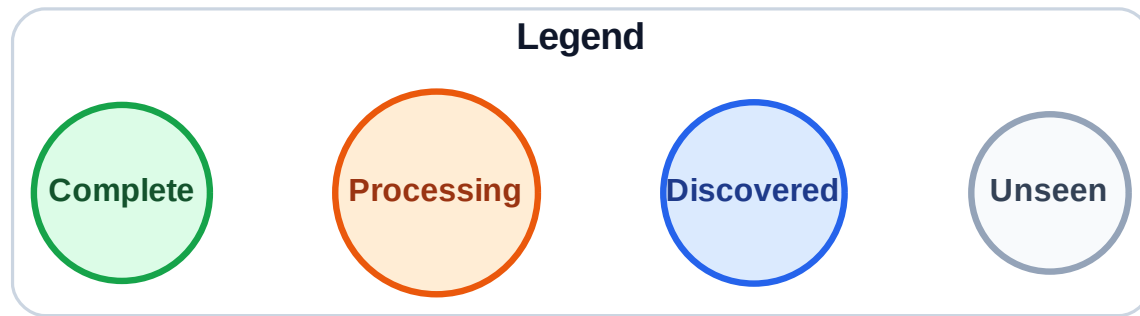


BFS Traversal

Step 6: Process node H

Queue: I → J → C → G

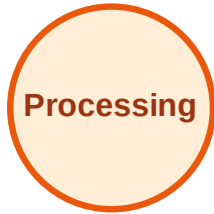
Visited: D, E



BFS Traversal

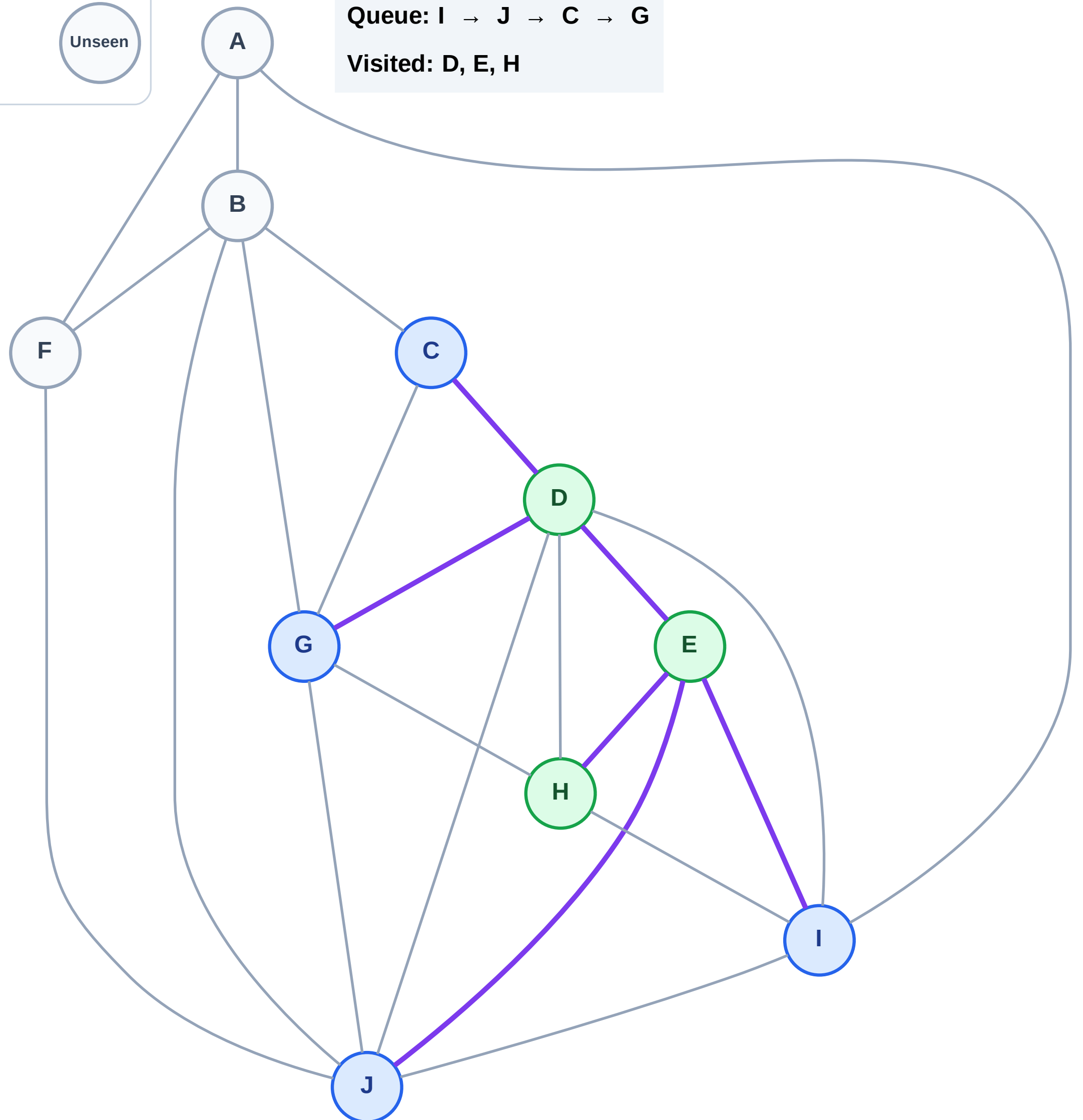
Step 7: No new neighbors from H

Legend



Queue: I → J → C → G

Visited: D, E, H



BFS Traversal

Step 8: Process node I

Legend

Complete

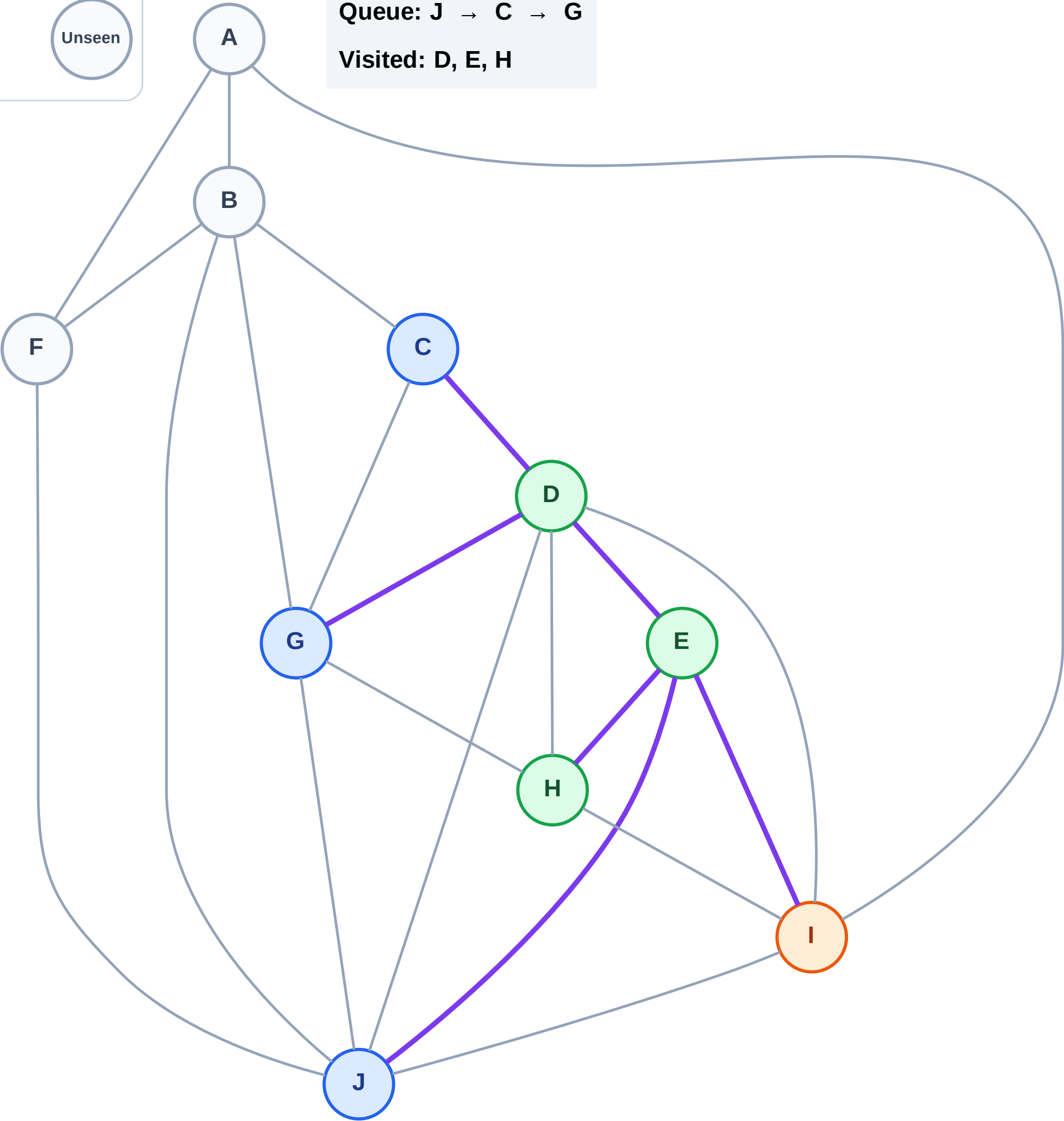
Processing

Discovered

Unseen

Queue: J → C → G

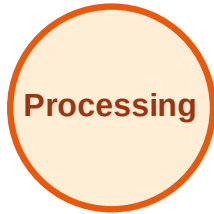
Visited: D, E, H



BFS Traversal

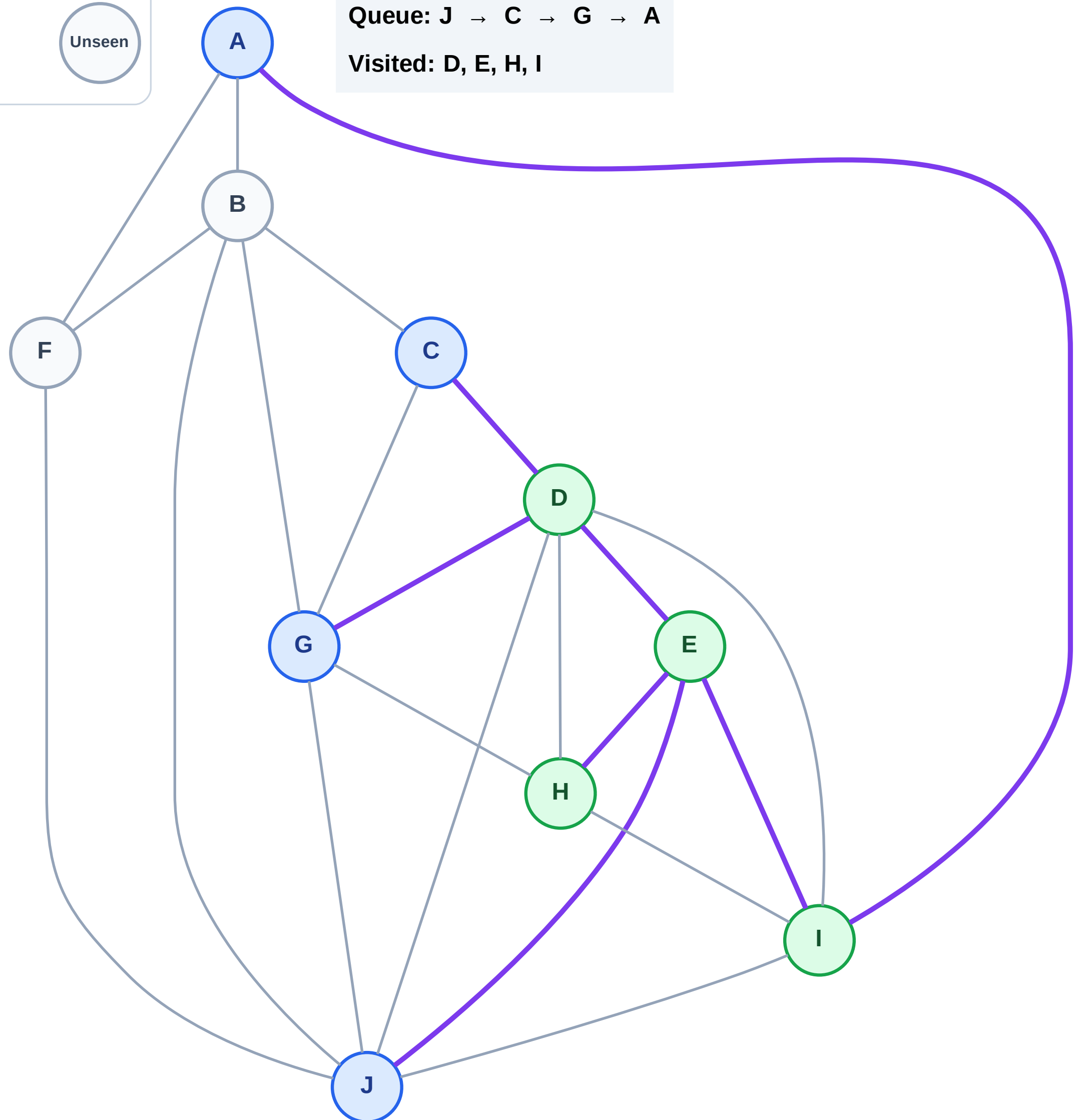
Step 9: Discover A from I

Legend



Queue: J → C → G → A

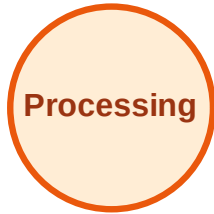
Visited: D, E, H, I



BFS Traversal

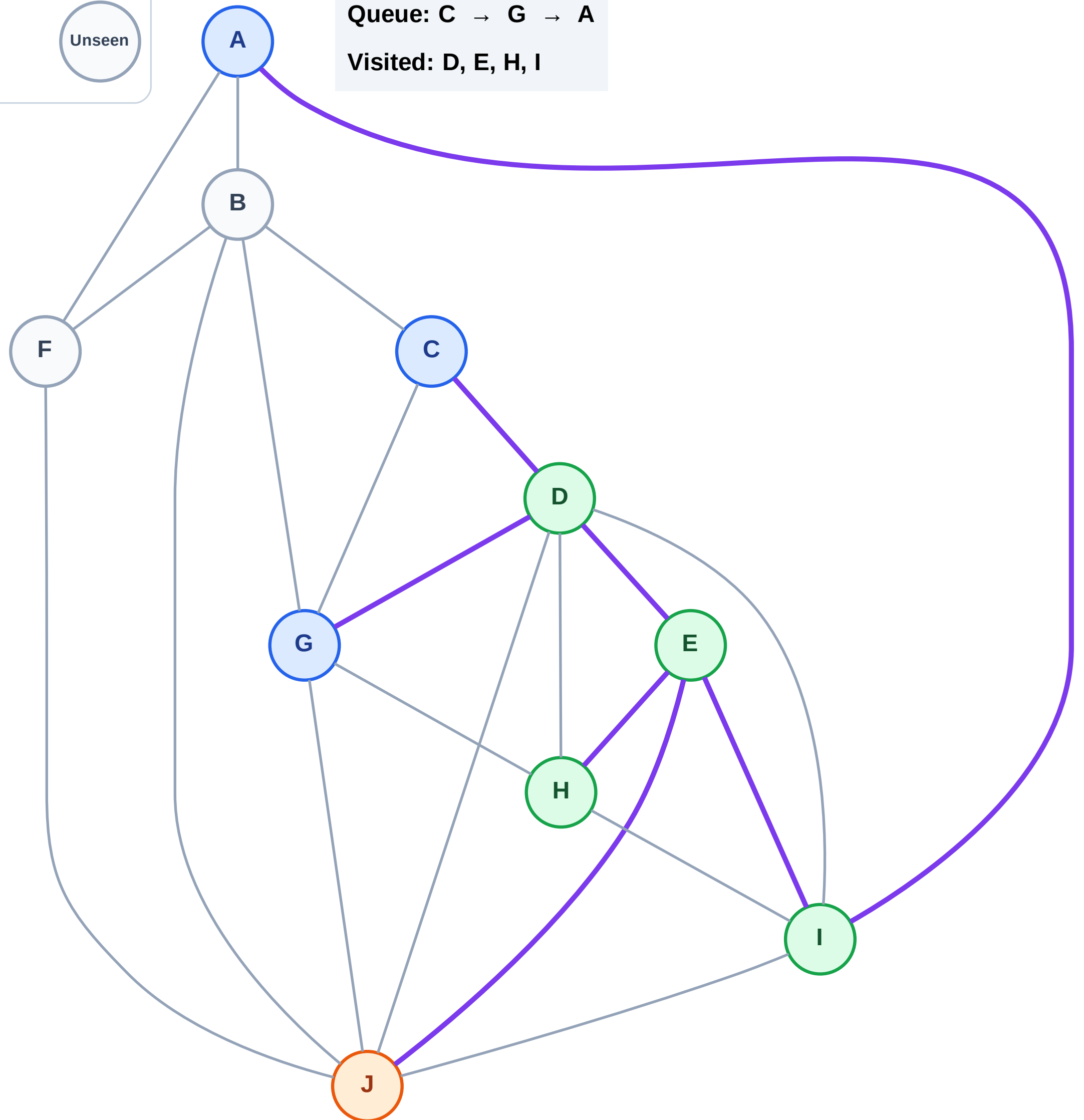
Step 10: Process node J

Legend



Queue: C → G → A

Visited: D, E, H, I



BFS Traversal

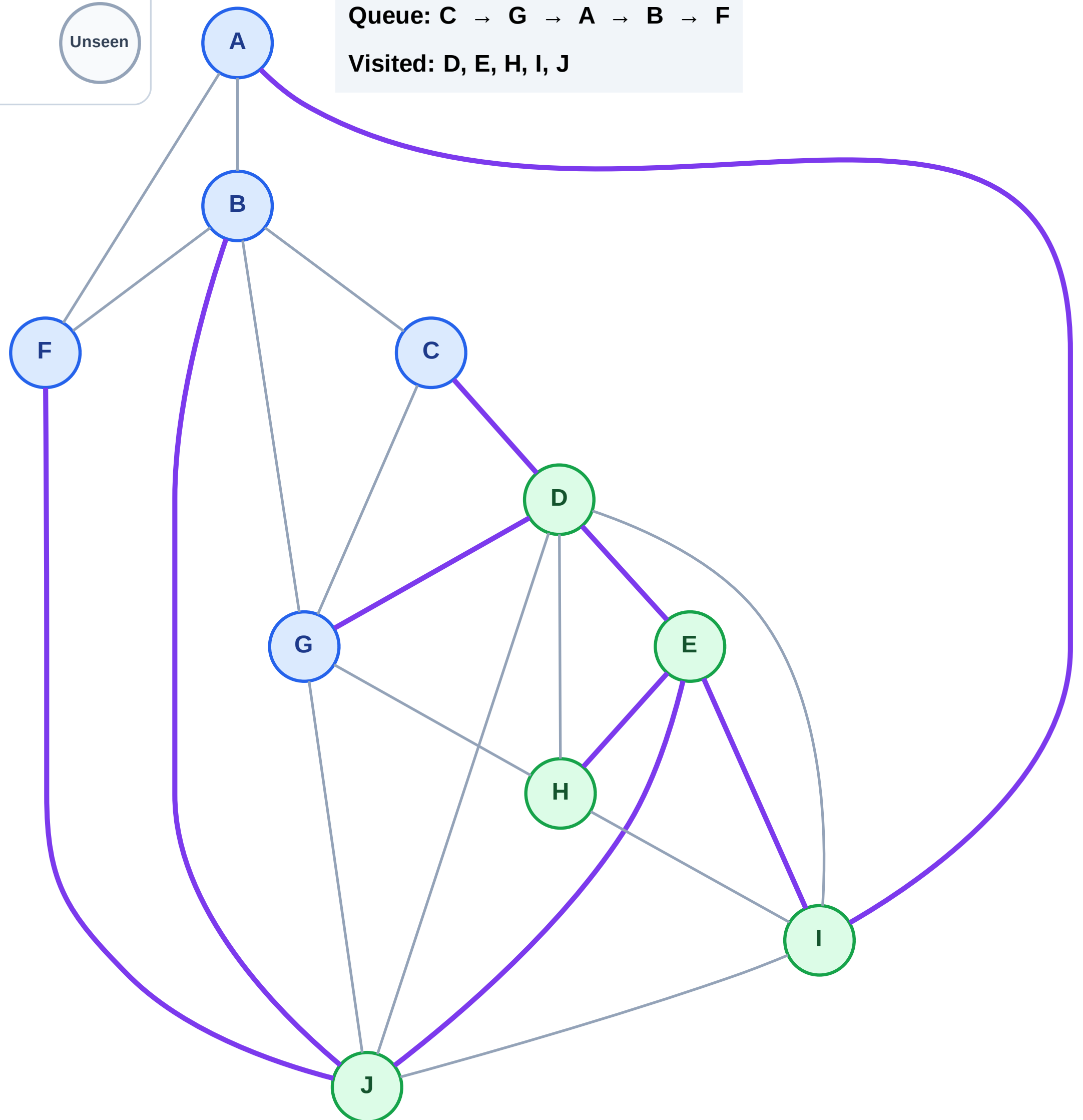
Step 11: Discover B, F from J

Legend



Queue: C → G → A → B → F

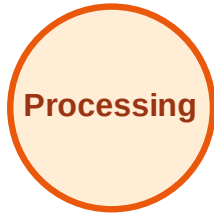
Visited: D, E, H, I, J



BFS Traversal

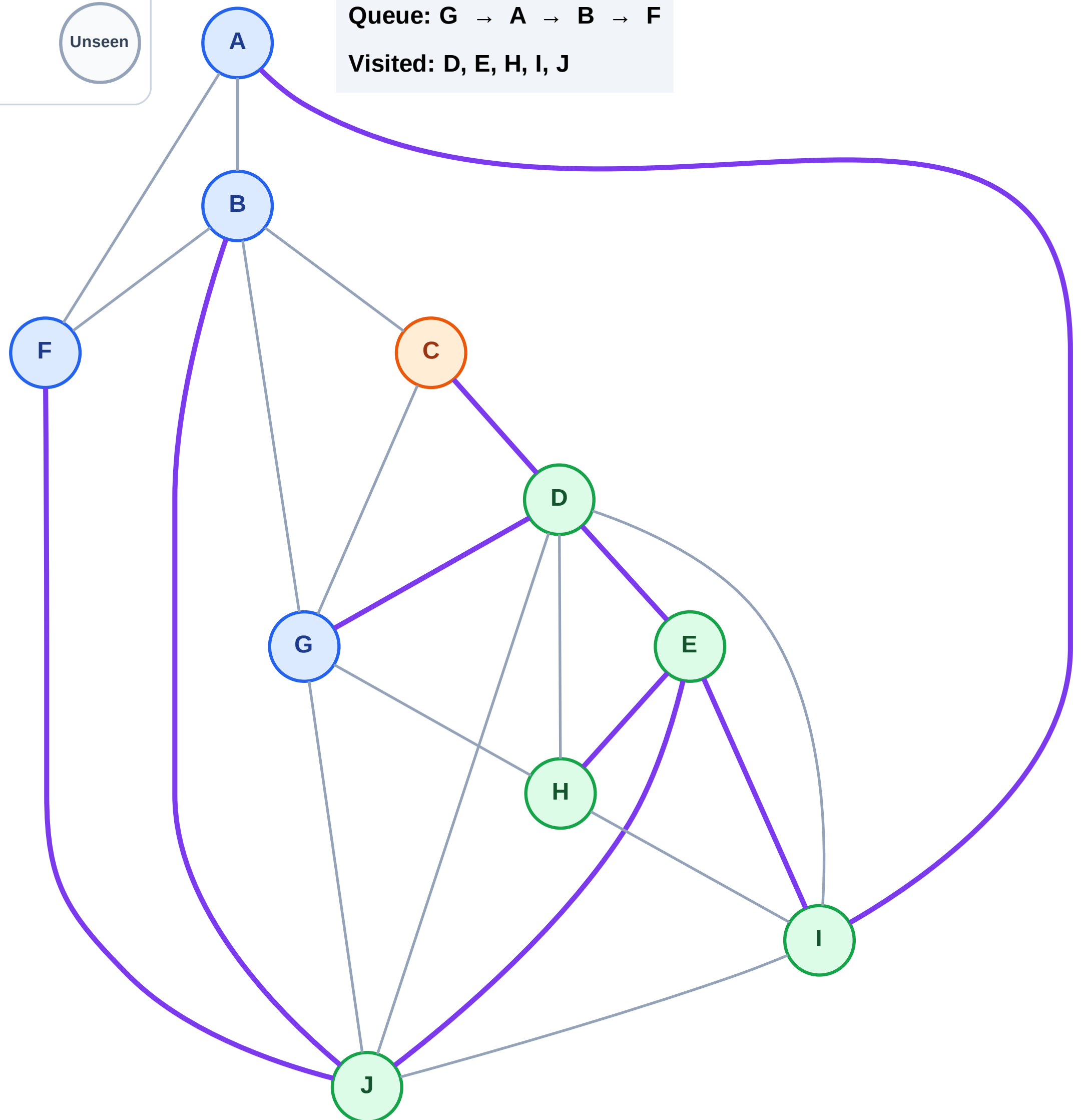
Step 12: Process node C

Legend



Queue: G → A → B → F

Visited: D, E, H, I, J

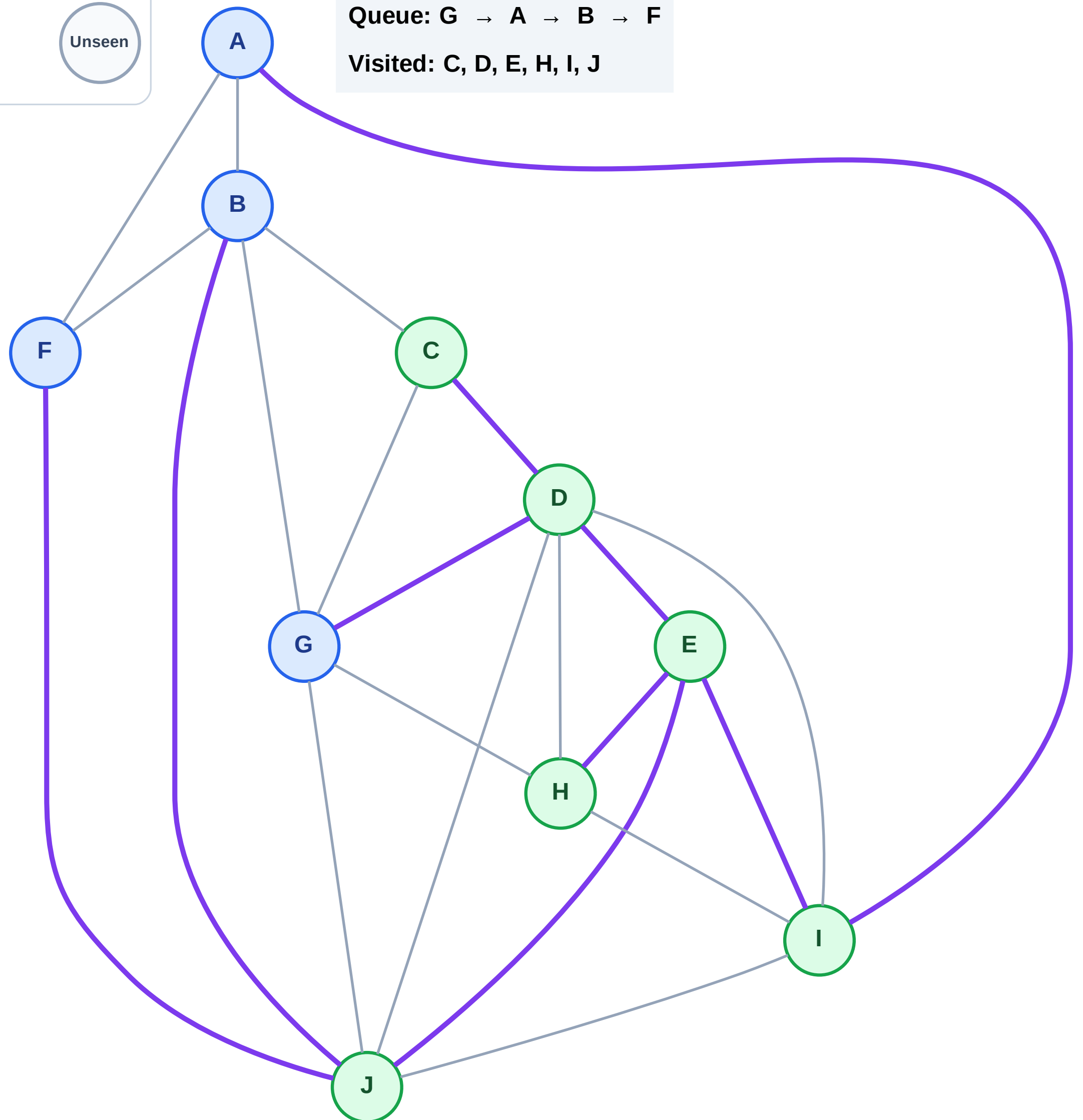


Step 13: No new neighbors from C

Queue: G → A → B → F

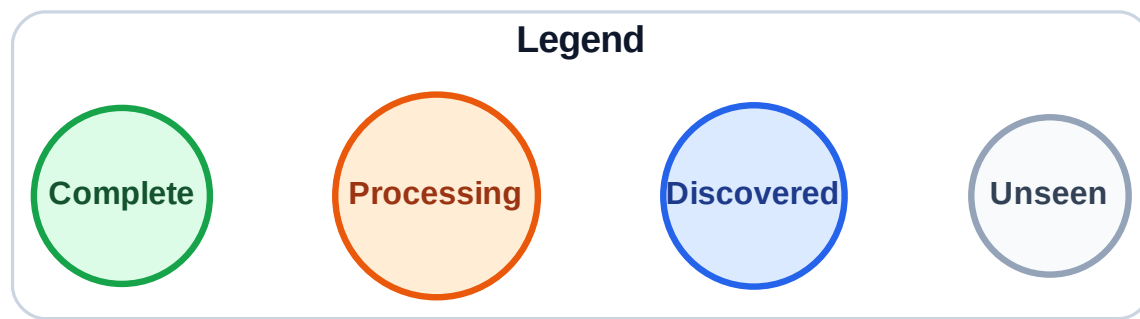
Legend

Unseen

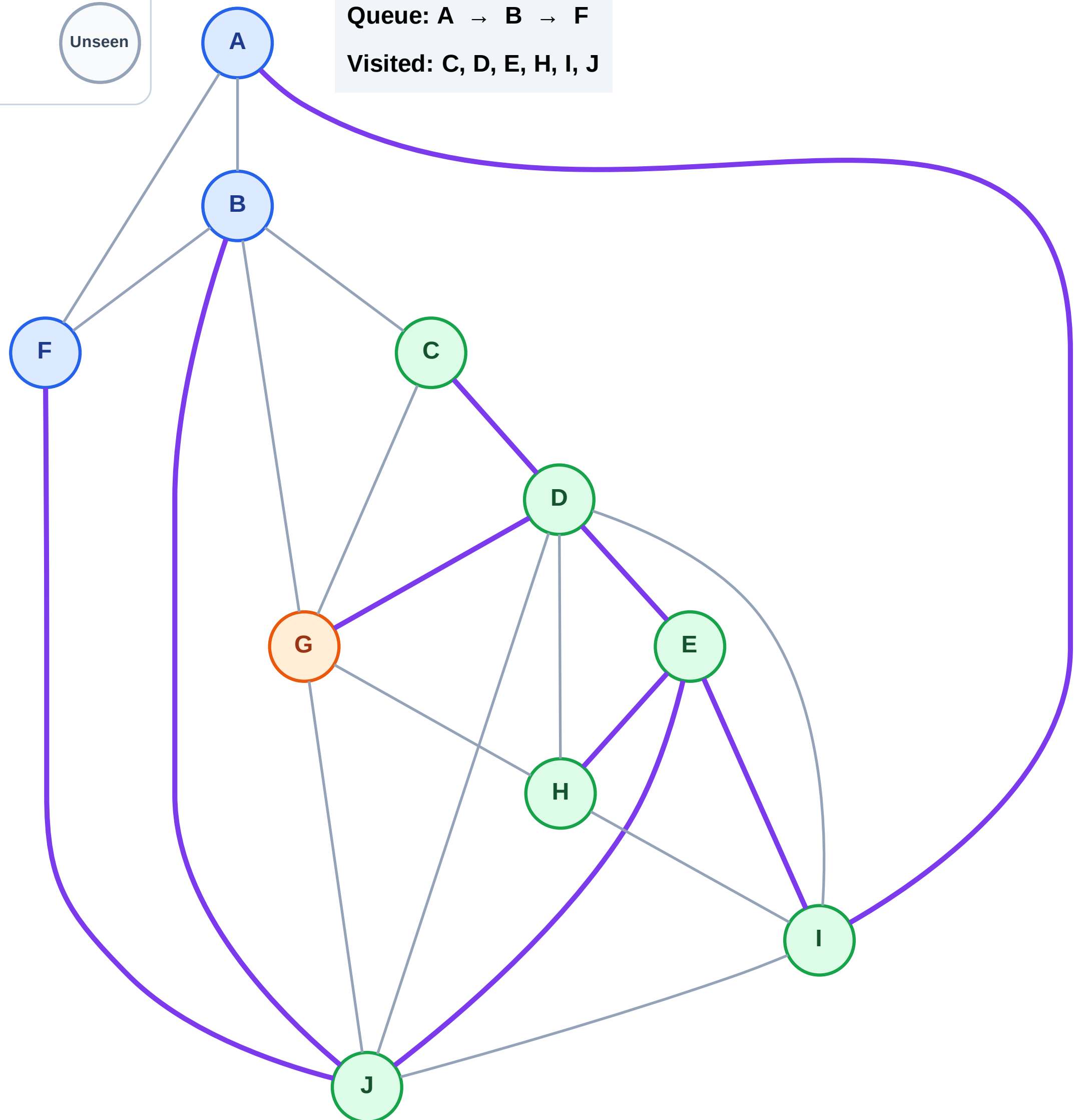


BFS Traversal

Step 14: Process node G

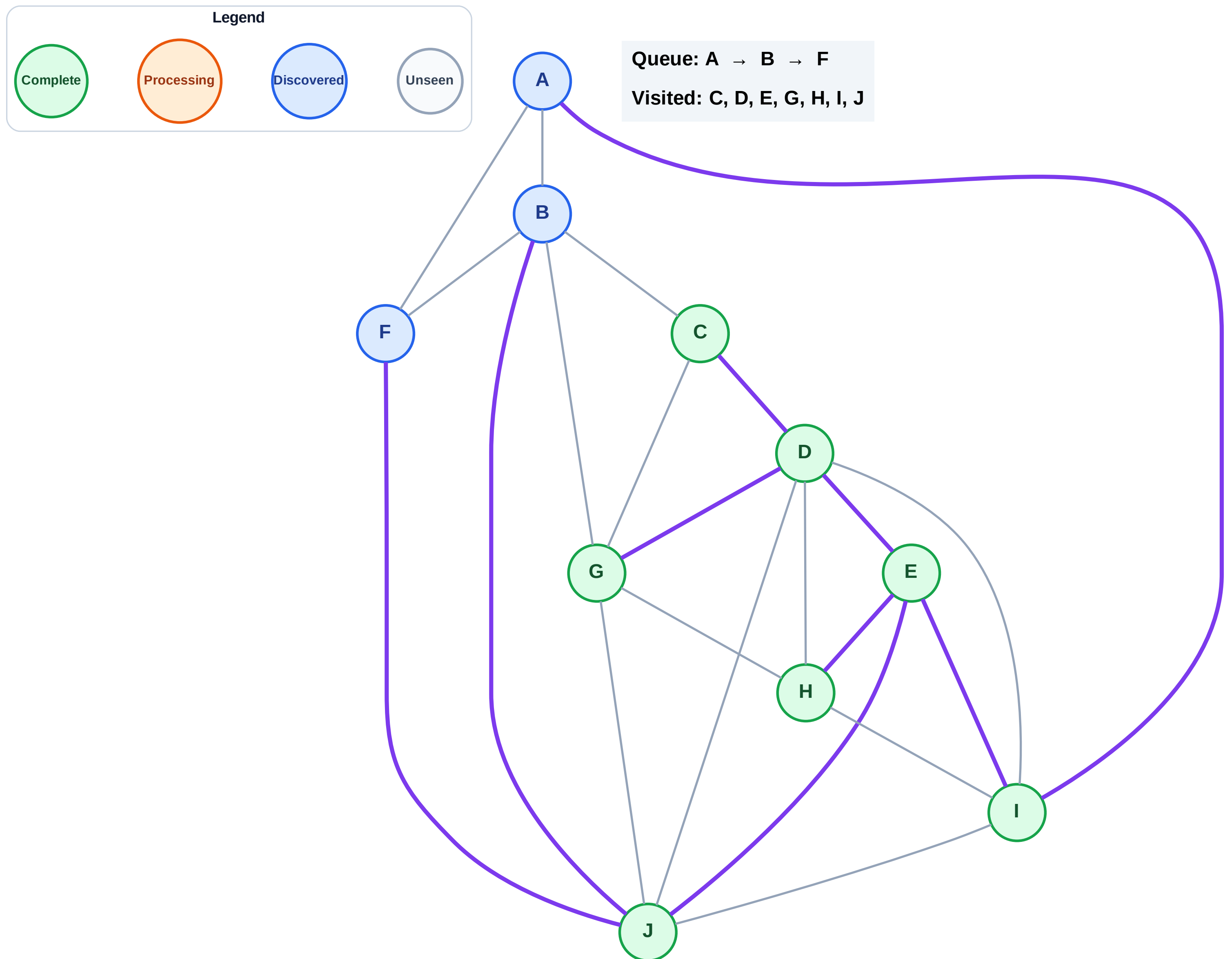


Queue: A → B → F
Visited: C, D, E, H, I, J



BFS Traversal

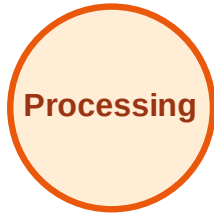
Step 15: No new neighbors from G



BFS Traversal

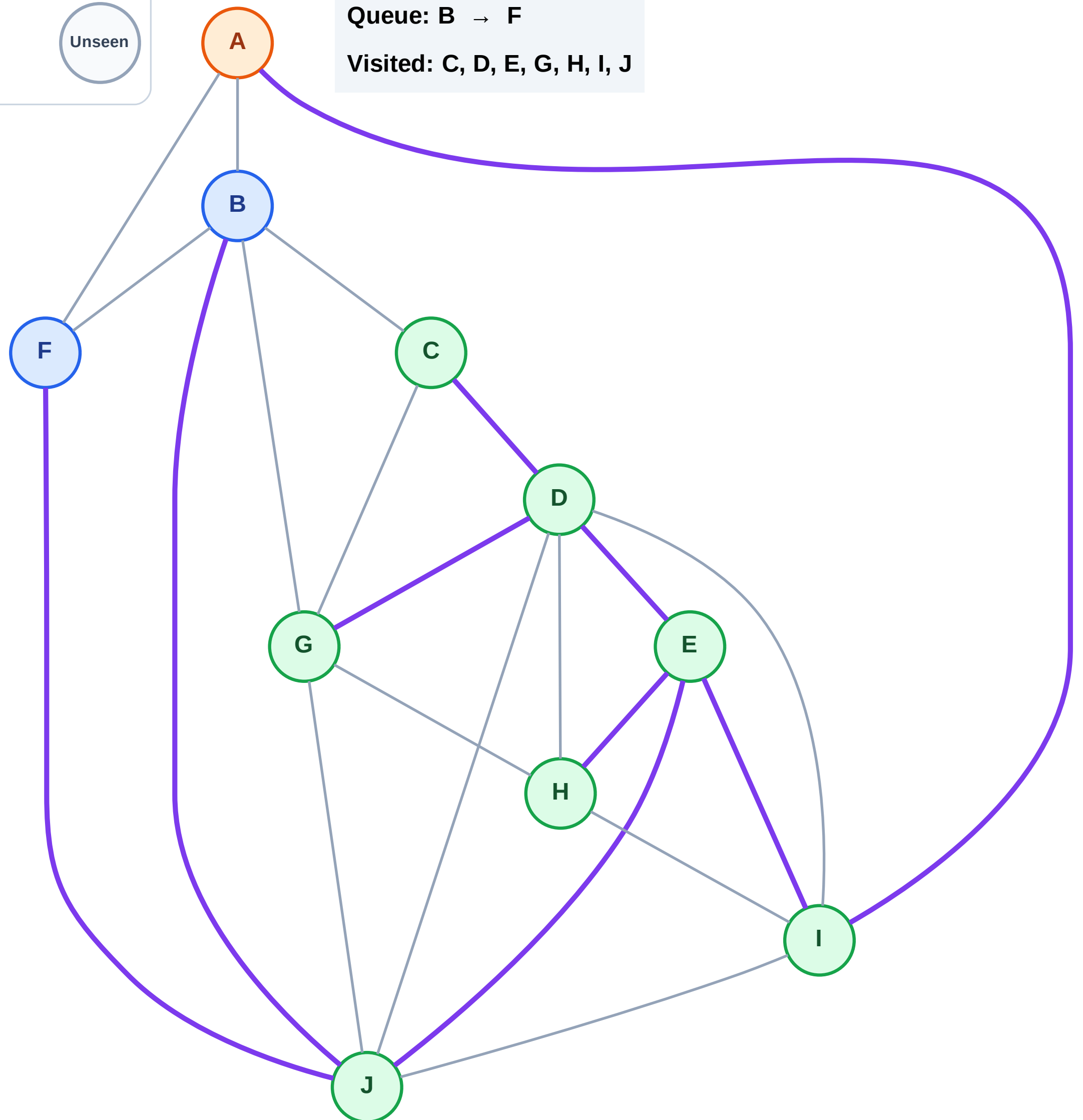
Step 16: Process node A

Legend



Queue: B → F

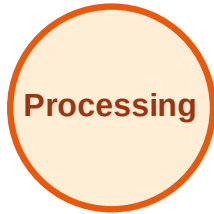
Visited: C, D, E, G, H, I, J



BFS Traversal

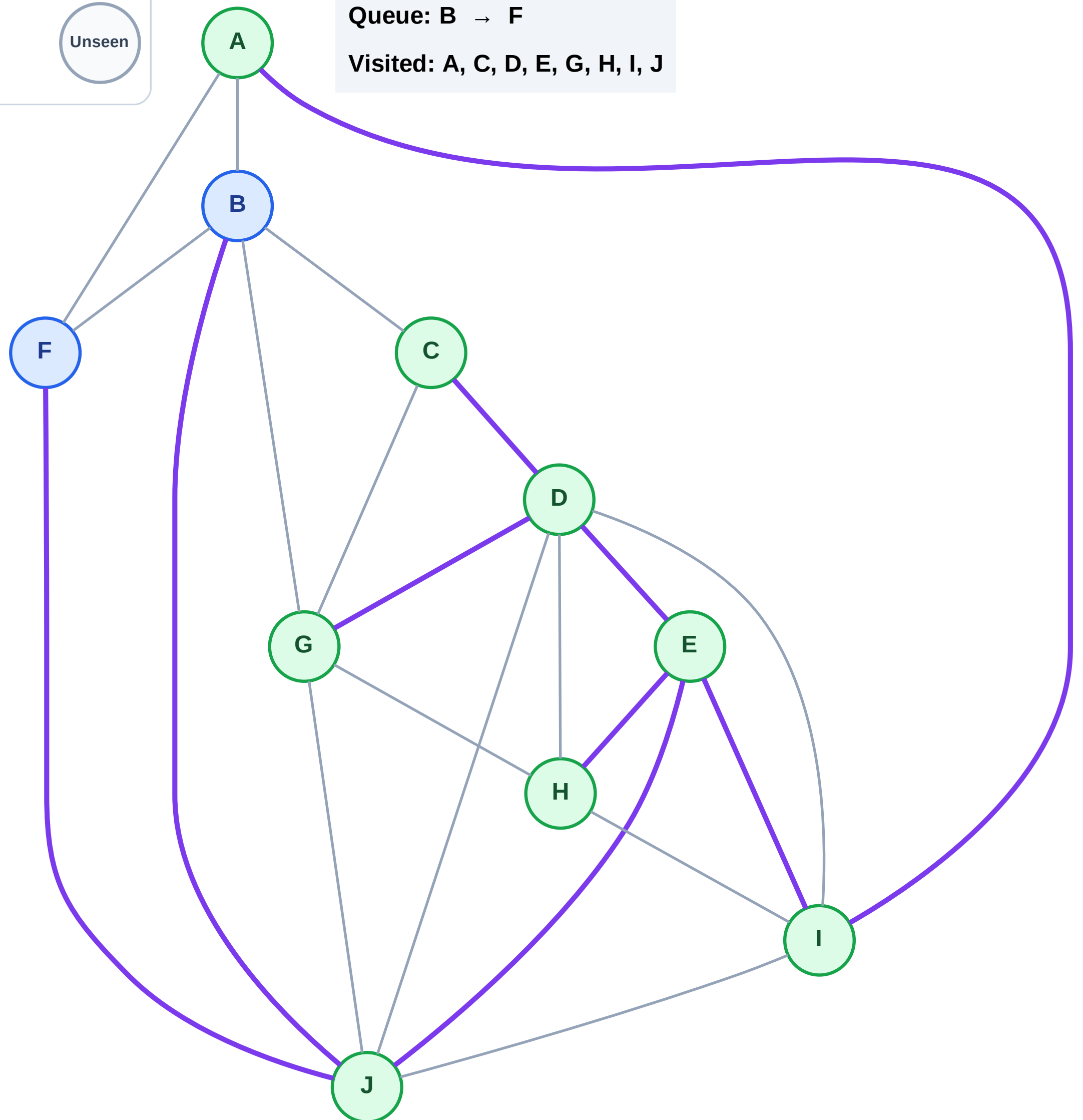
Step 17: No new neighbors from A

Legend



Queue: B → F

Visited: A, C, D, E, G, H, I, J



BFS Traversal

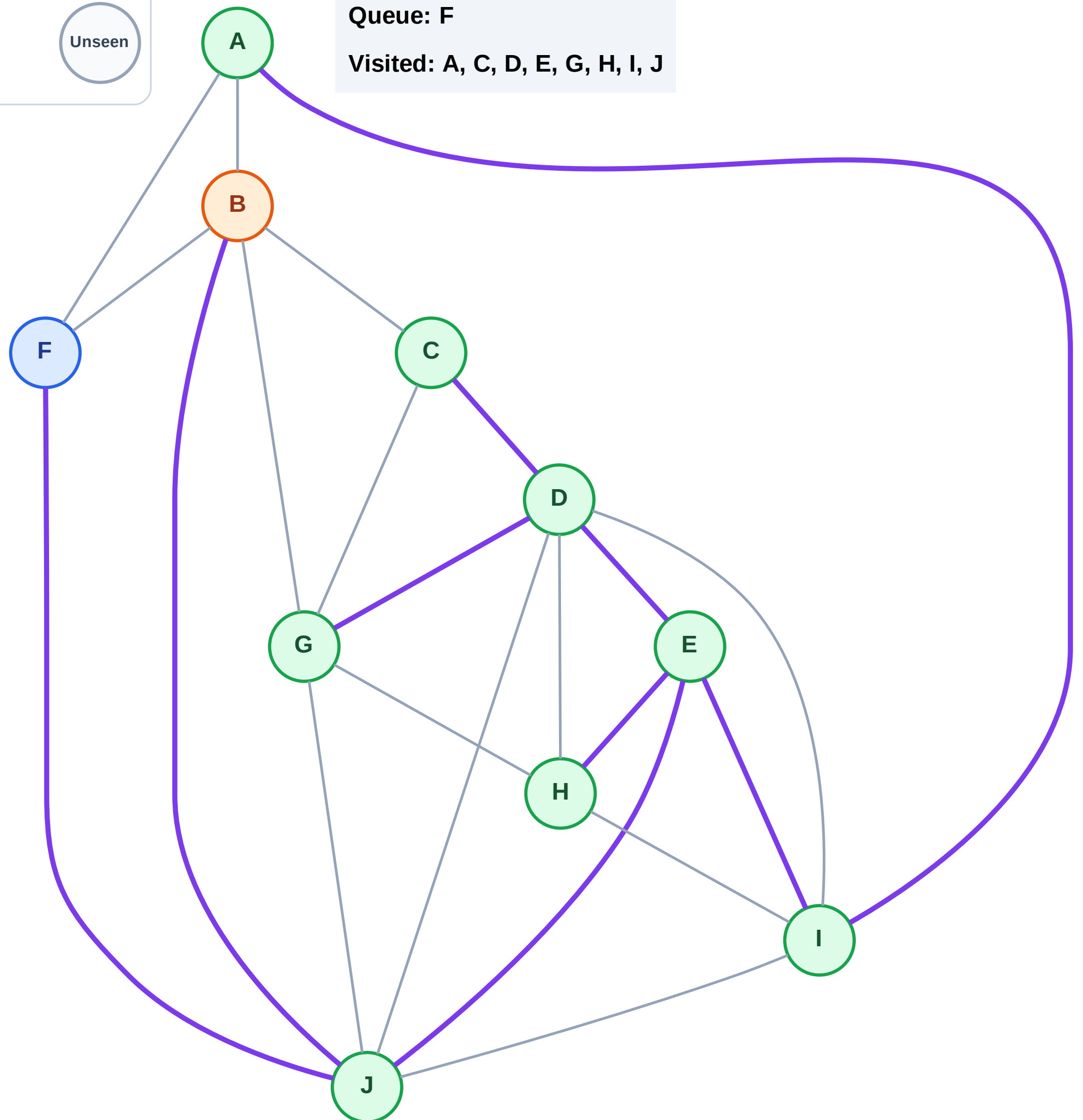
Step 18: Process node B

Legend



Queue: F

Visited: A, C, D, E, G, H, I, J



BFS Traversal

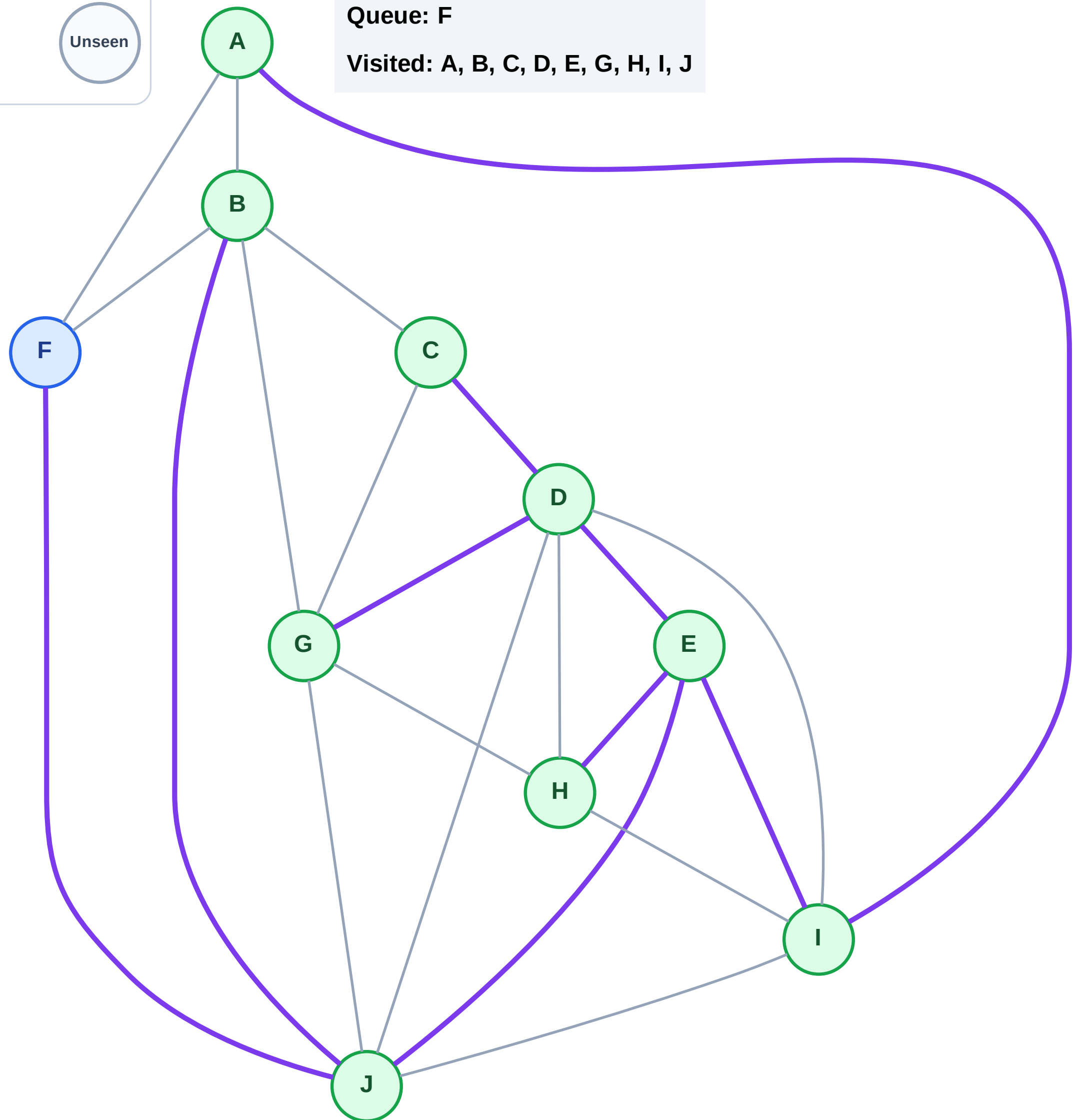
Step 19: No new neighbors from B

Legend



Queue: F

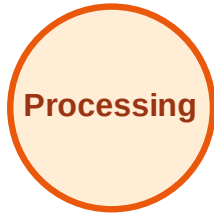
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

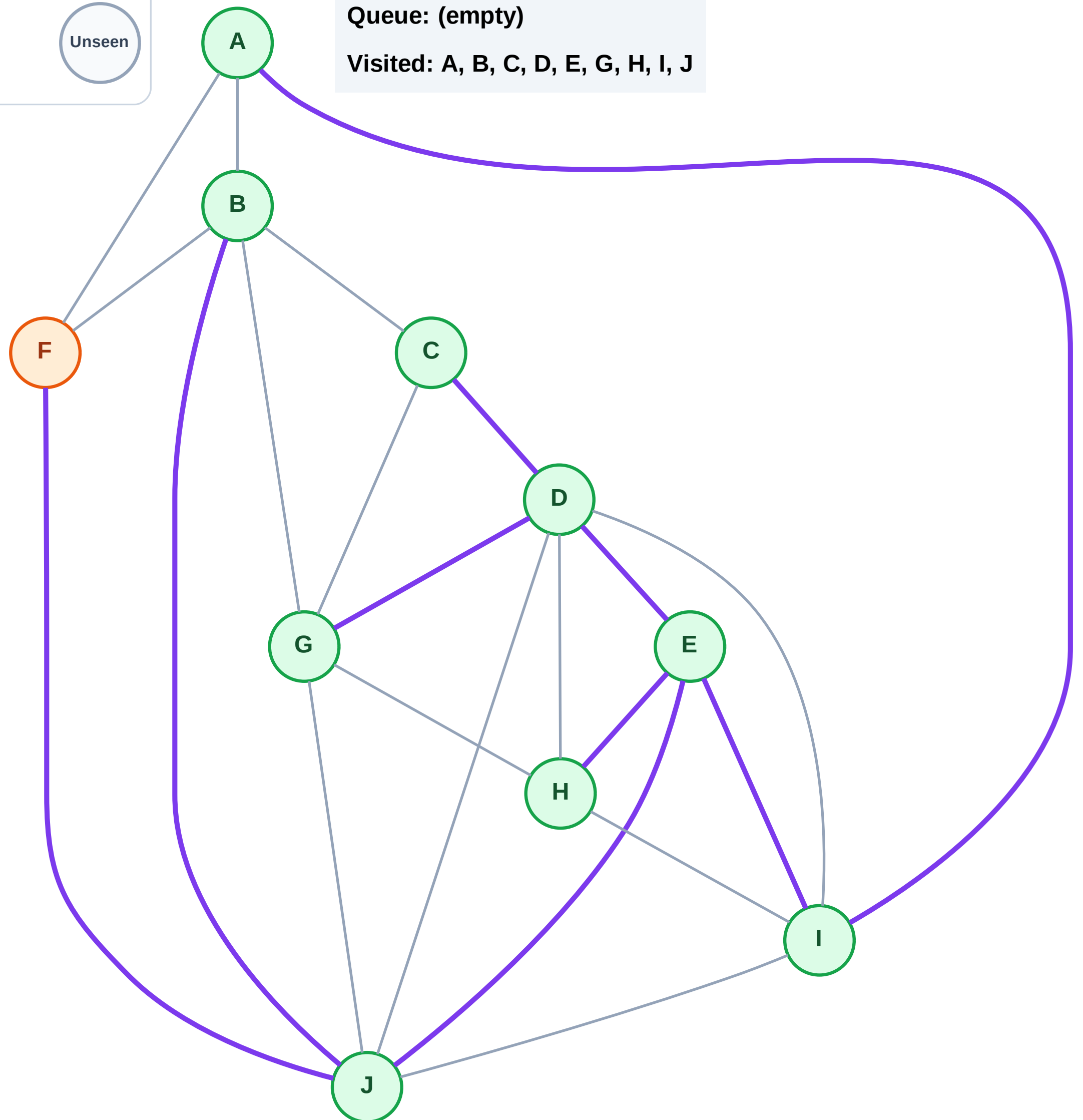
Step 20: Process node F

Legend



Queue: (empty)

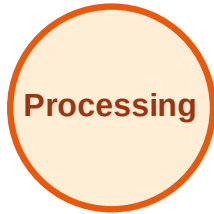
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

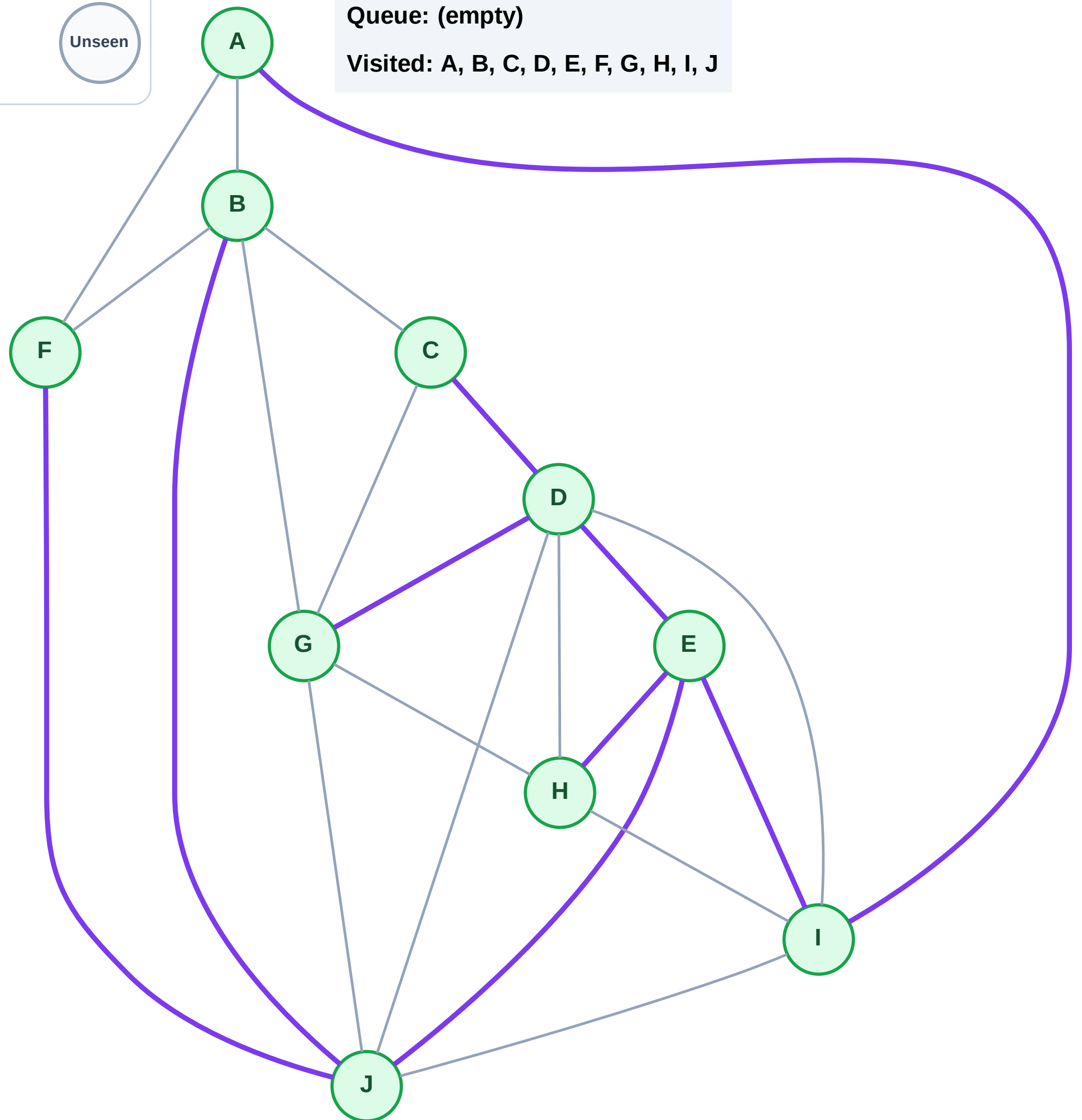
Step 21: No new neighbors from F

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

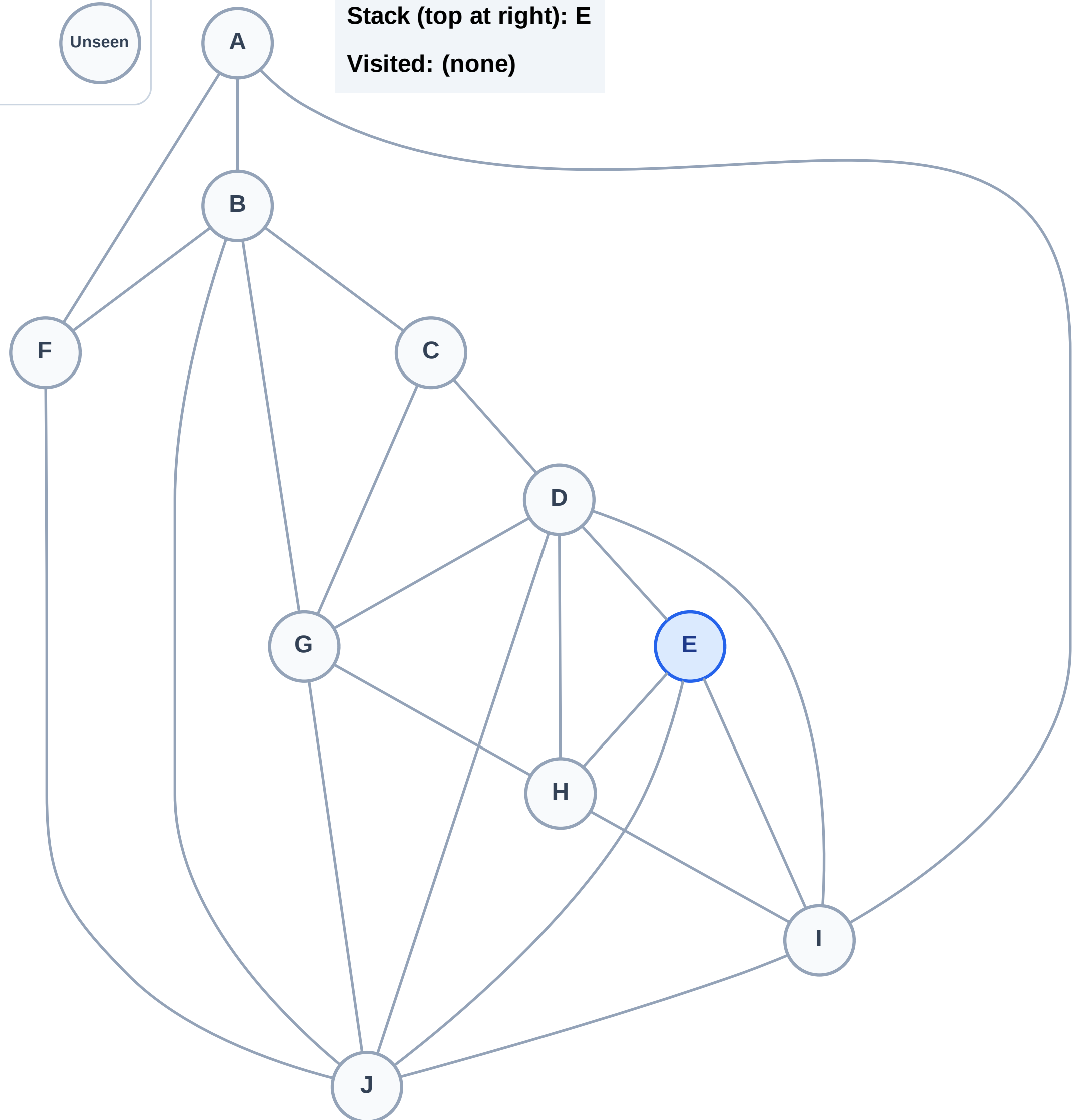
Step 1: Start at E

Legend



Stack (top at right): E

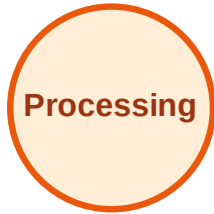
Visited: (none)



DFS Traversal

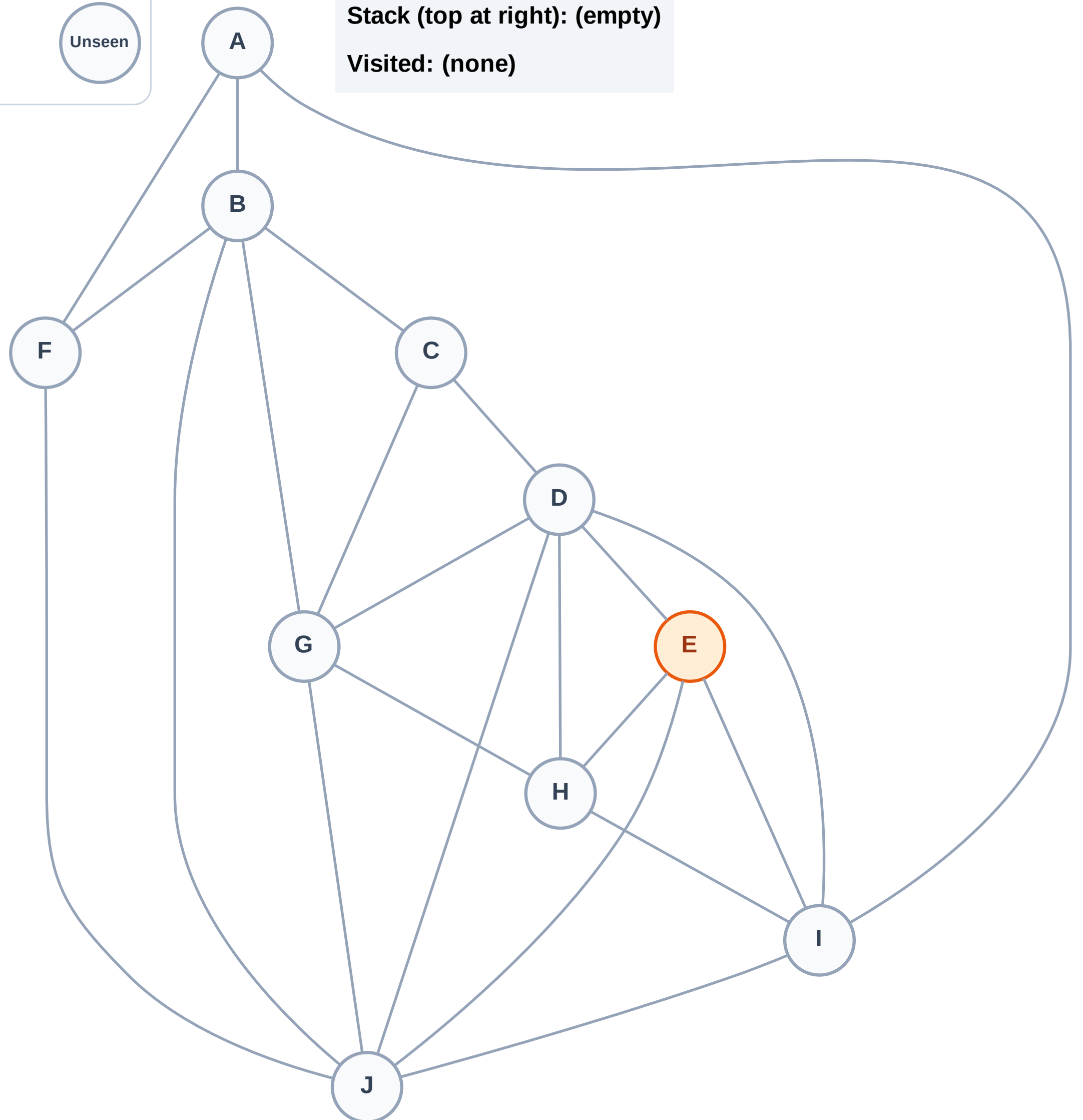
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

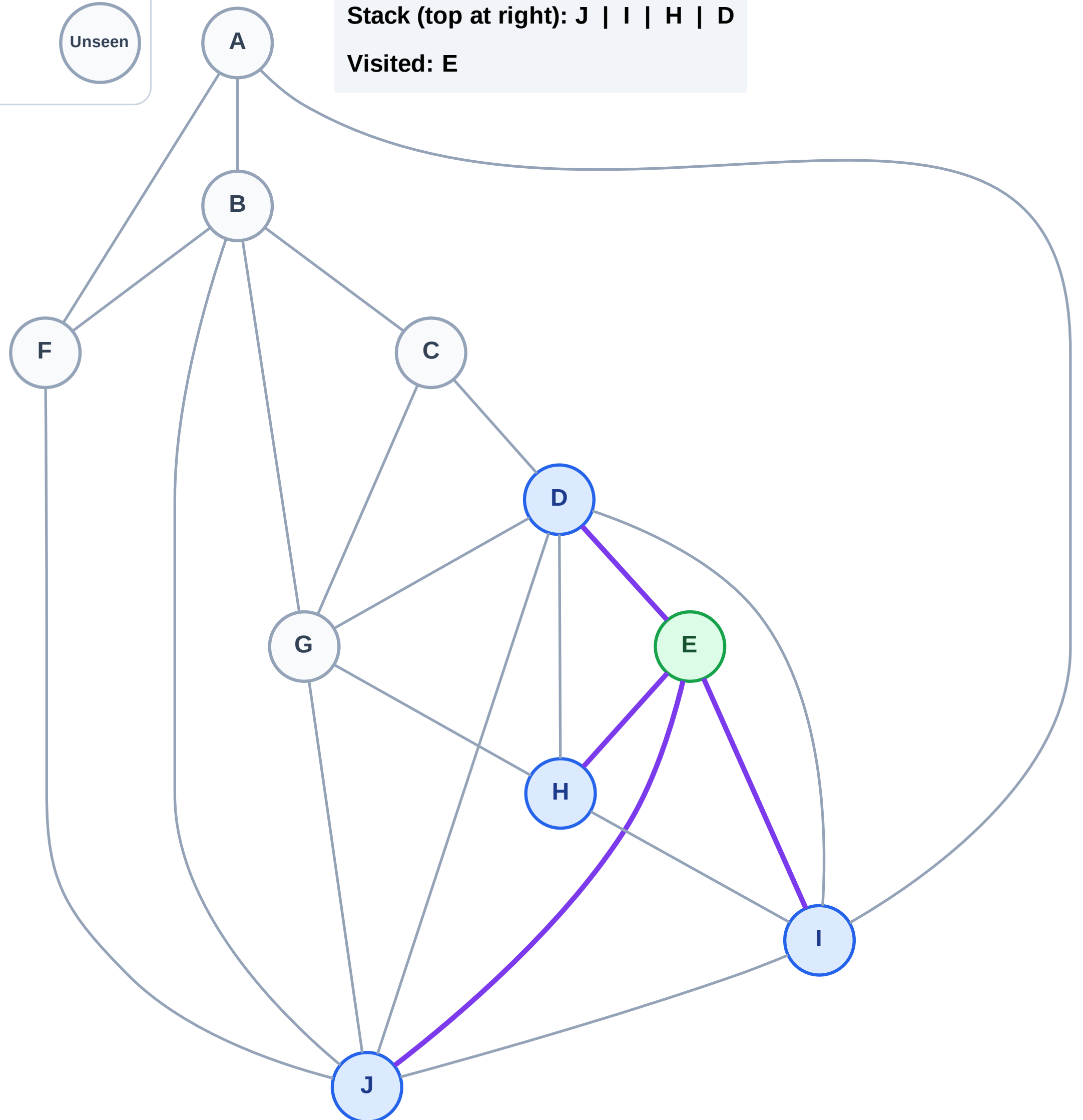
Step 3: Discover J, I, H, D from E

Legend



Stack (top at right): J | I | H | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend

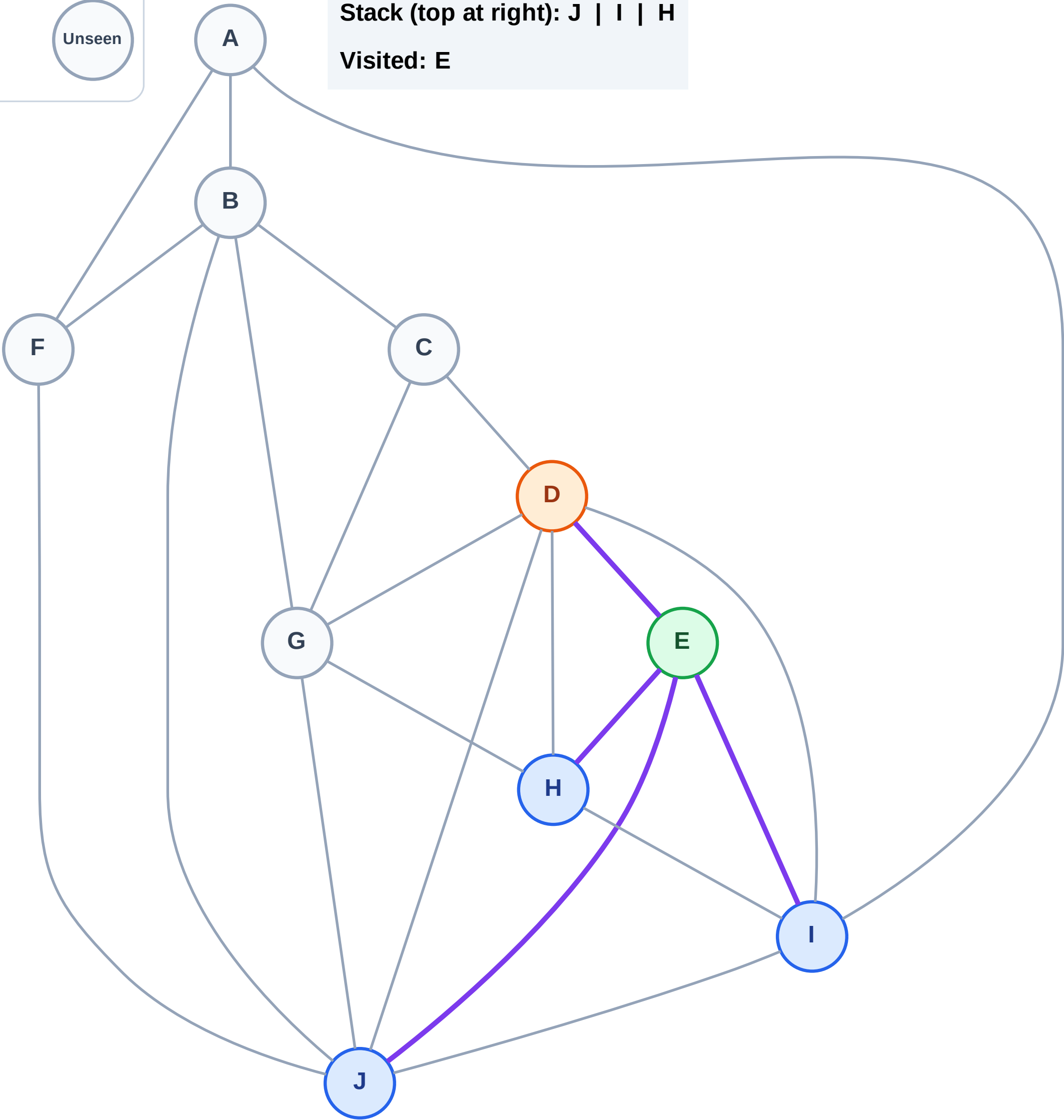
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | H
Visited: E



DFS Traversal

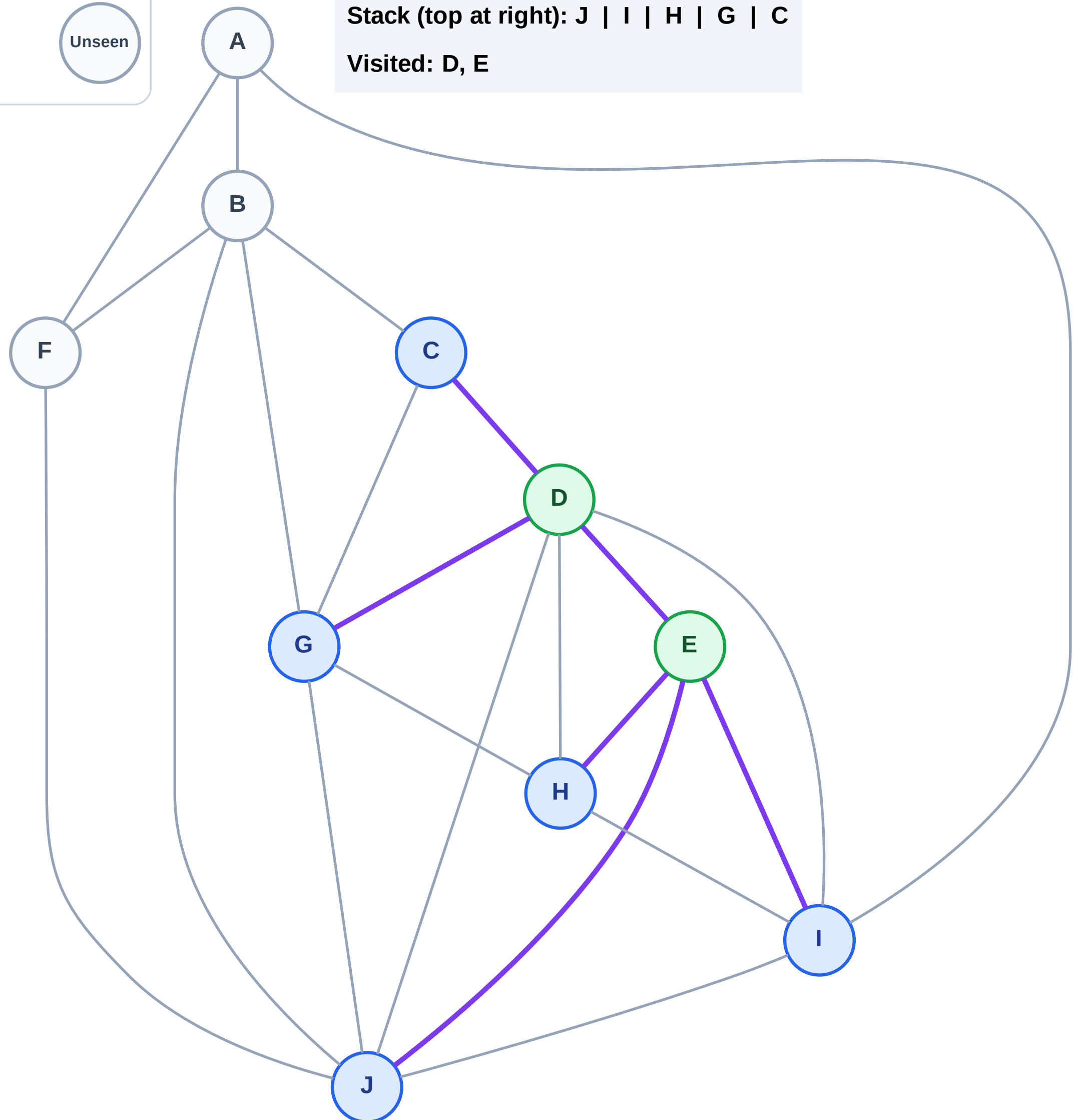
Step 5: Discover G, C from D

Legend

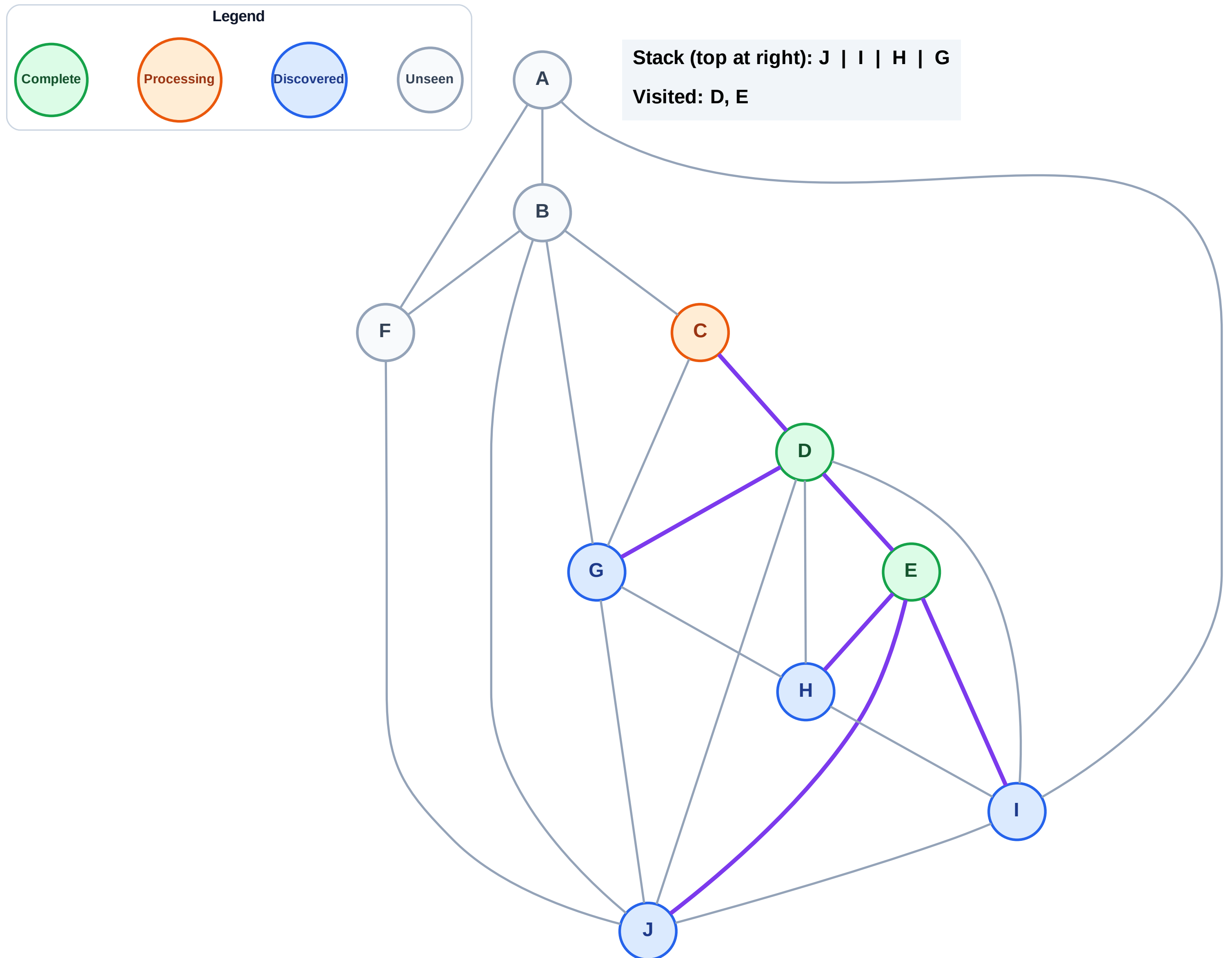


Stack (top at right): J | I | H | G | C

Visited: D, E



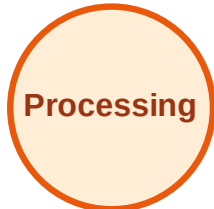
Step 6: Process node C



DFS Traversal

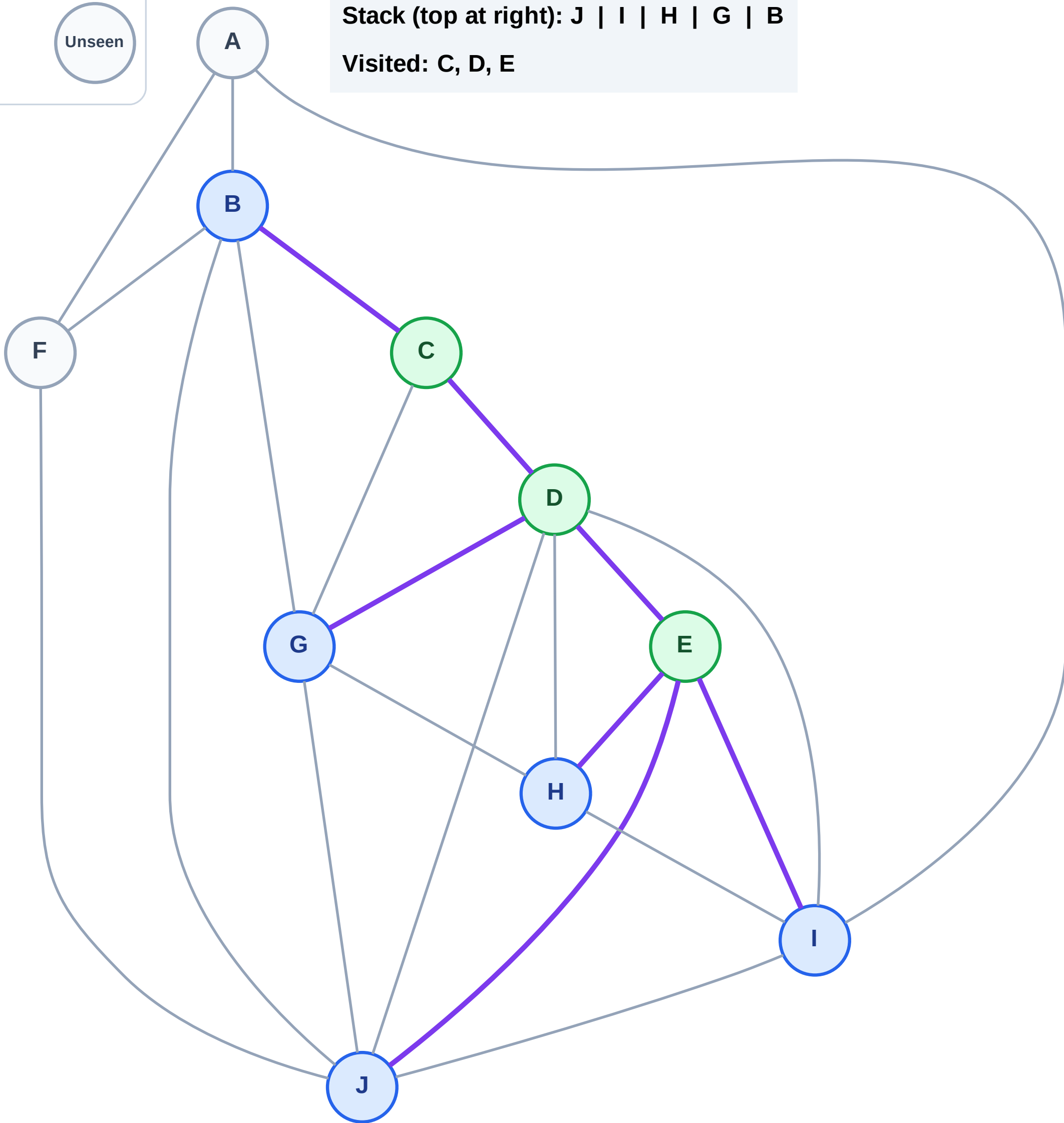
Step 7: Discover B from C

Legend



Stack (top at right): J | I | H | G | B

Visited: C, D, E



DFS Traversal

Step 8: Process node B

Legend

Complete

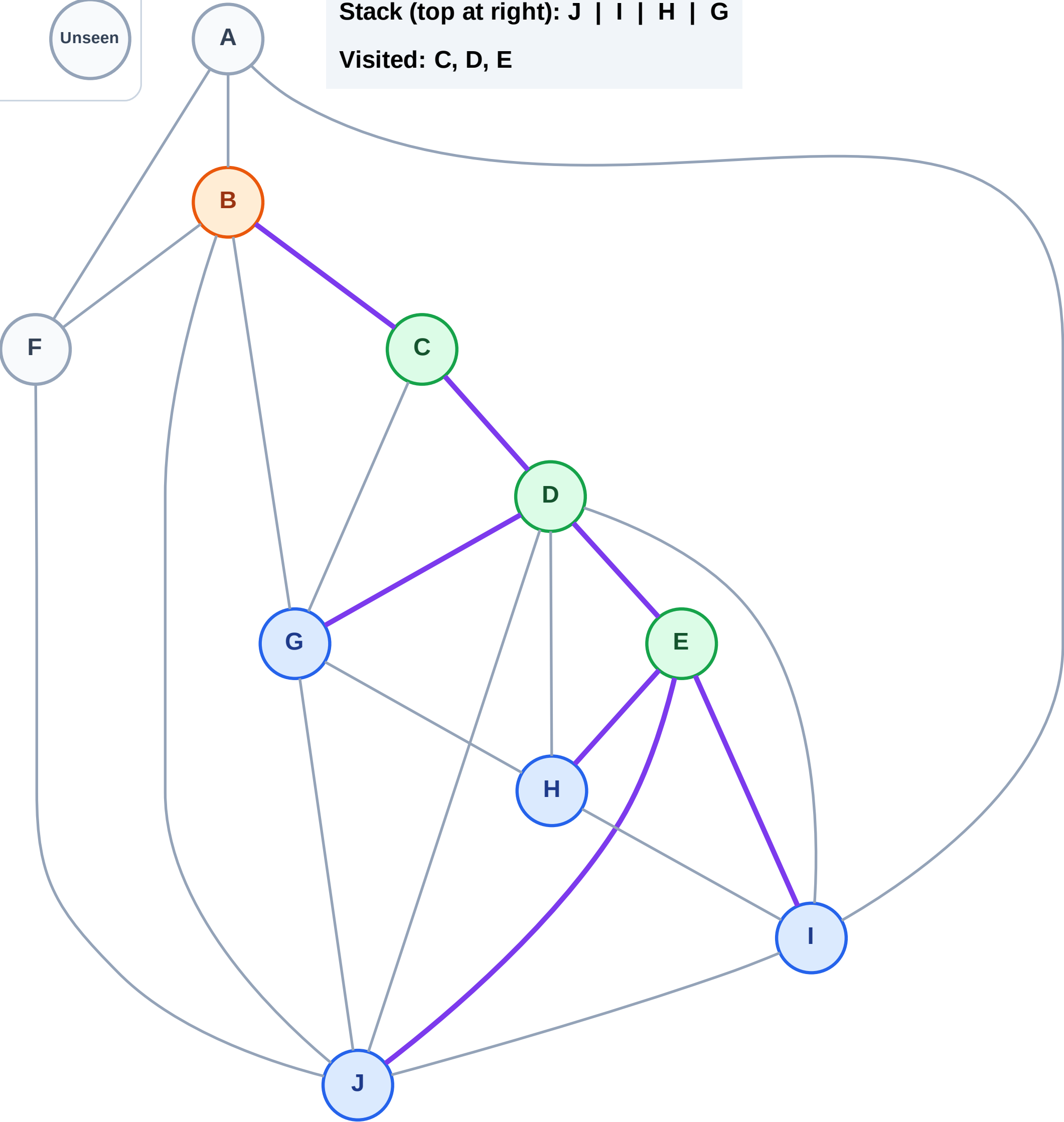
Processing

Discovered

Unseen

Stack (top at right): J | I | H | G

Visited: C, D, E



DFS Traversal

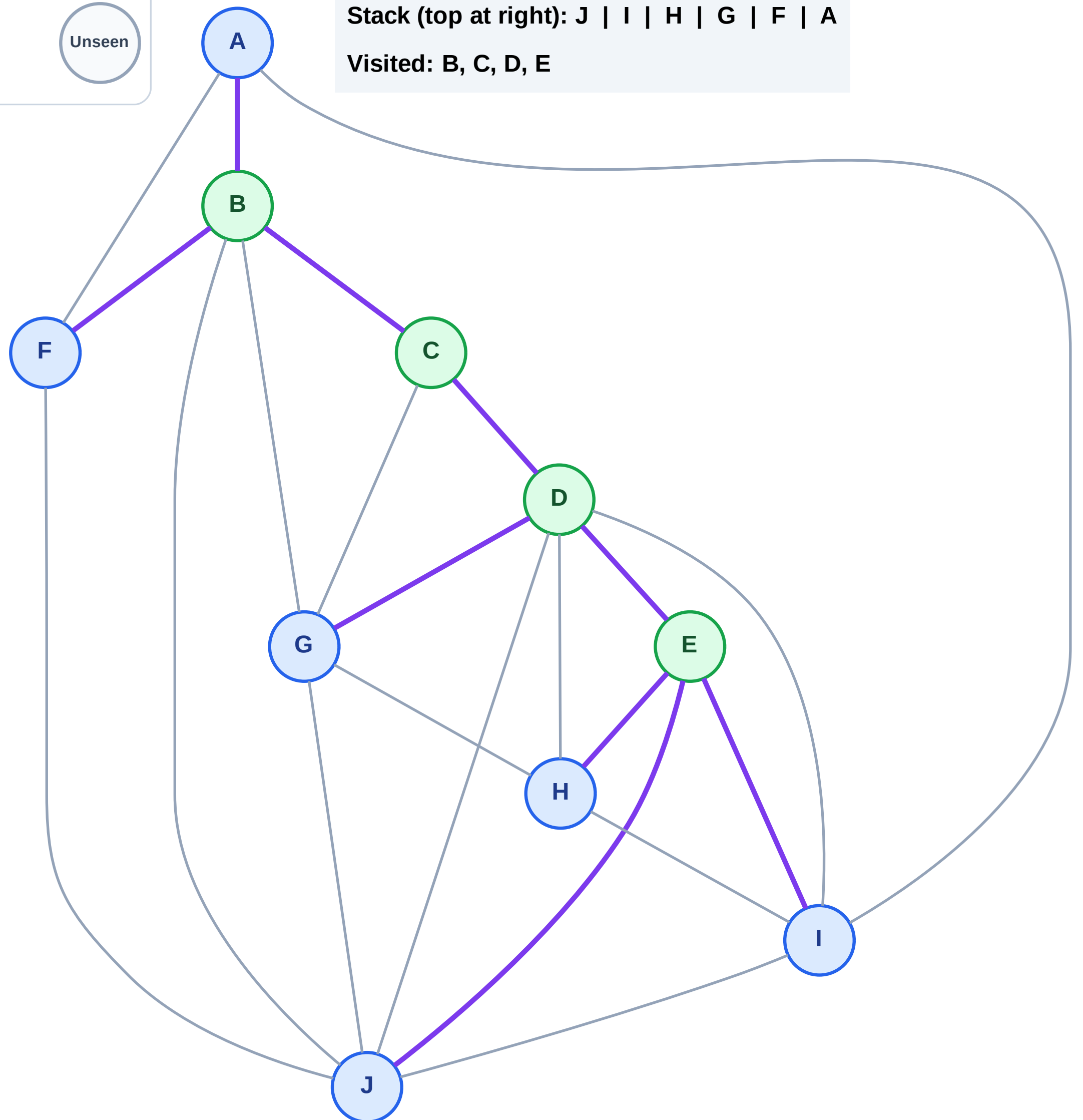
Step 9: Discover F, A from B

Legend



Stack (top at right): J | I | H | G | F | A

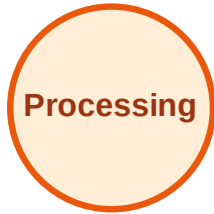
Visited: B, C, D, E



DFS Traversal

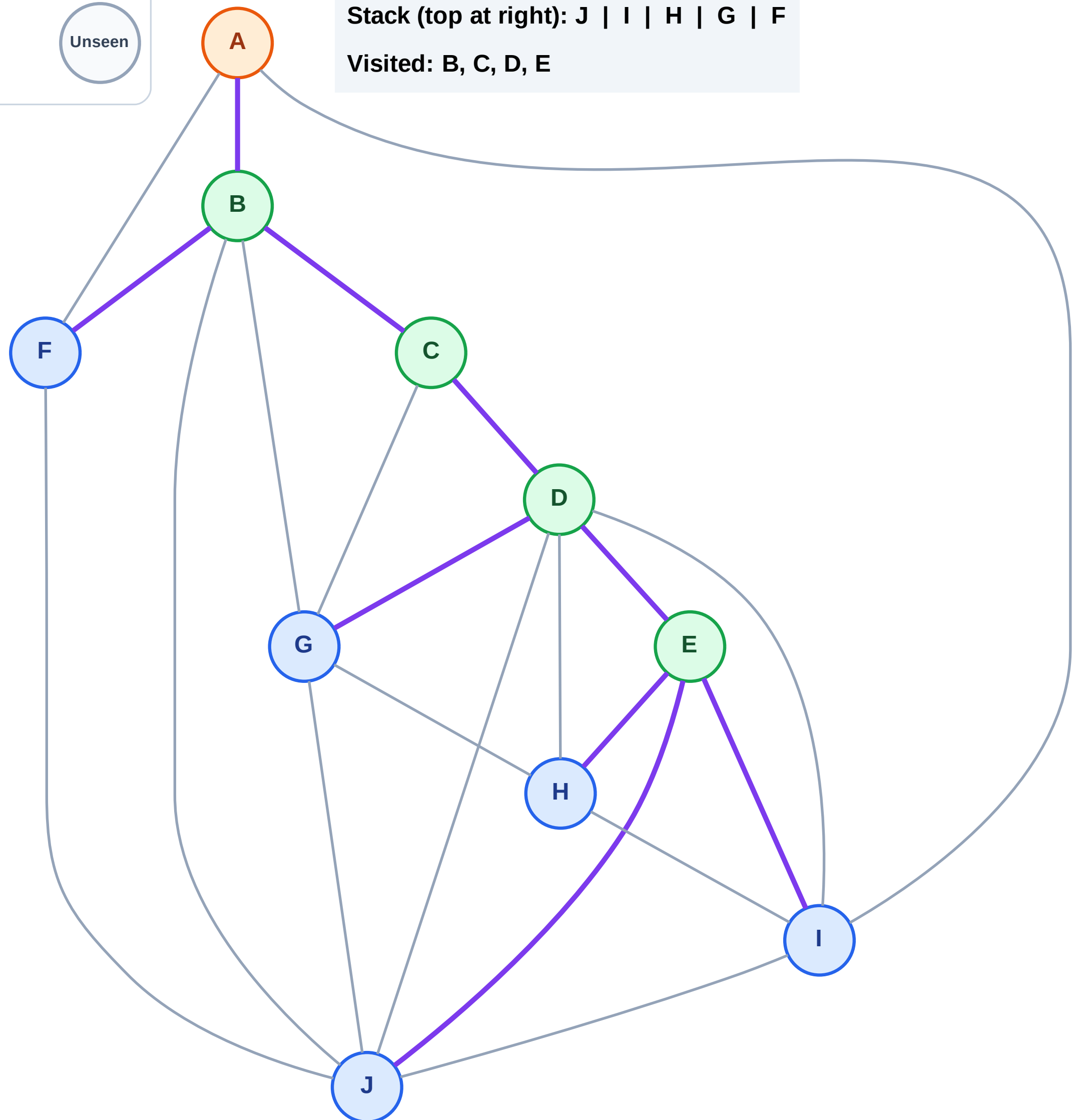
Step 10: Process node A

Legend



Stack (top at right): J | I | H | G | F

Visited: B, C, D, E



DFS Traversal

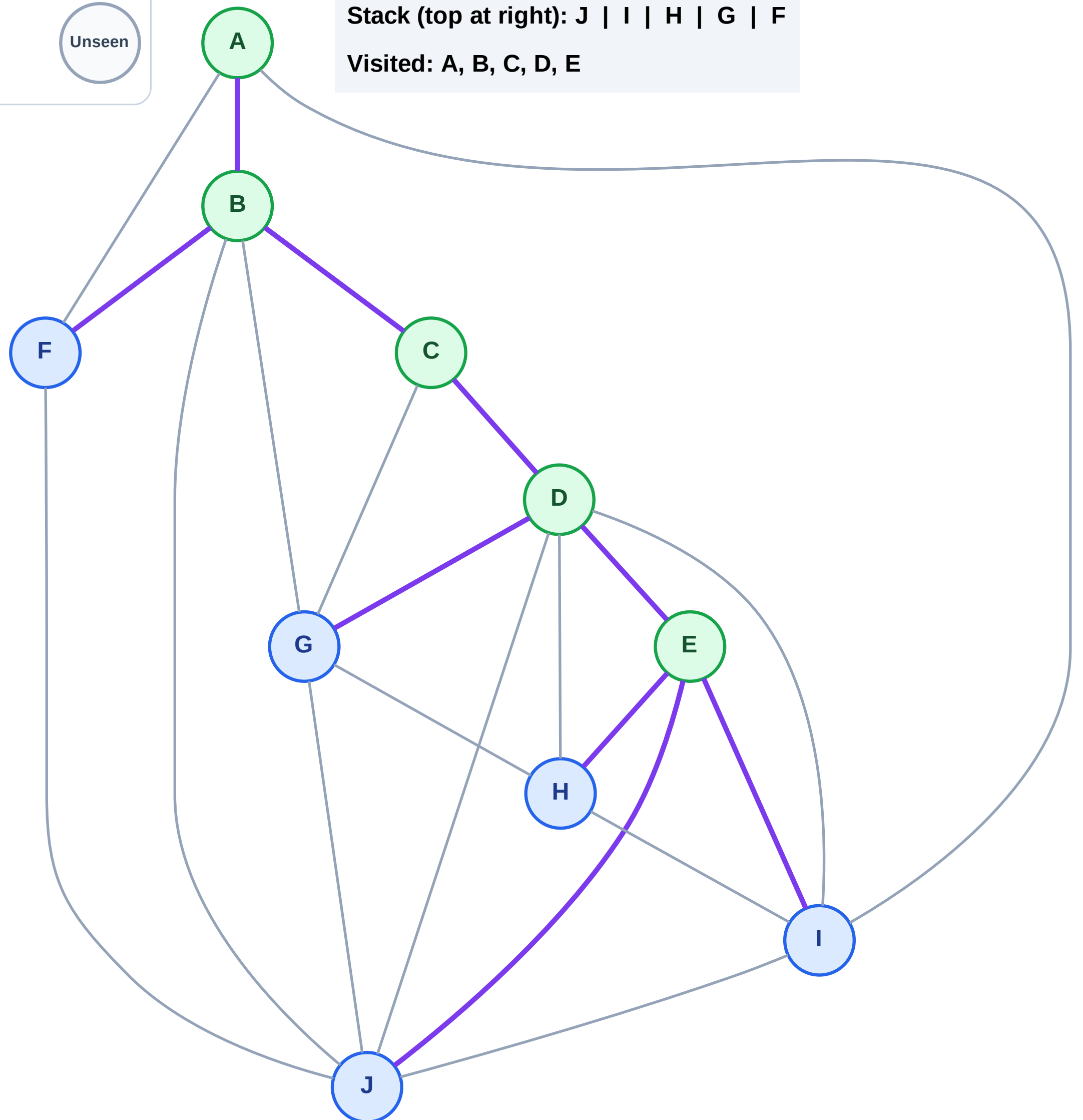
Step 11: No new neighbors from A

Legend



Stack (top at right): J | I | H | G | F

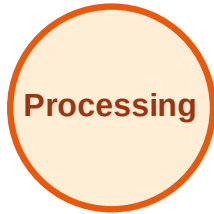
Visited: A, B, C, D, E



DFS Traversal

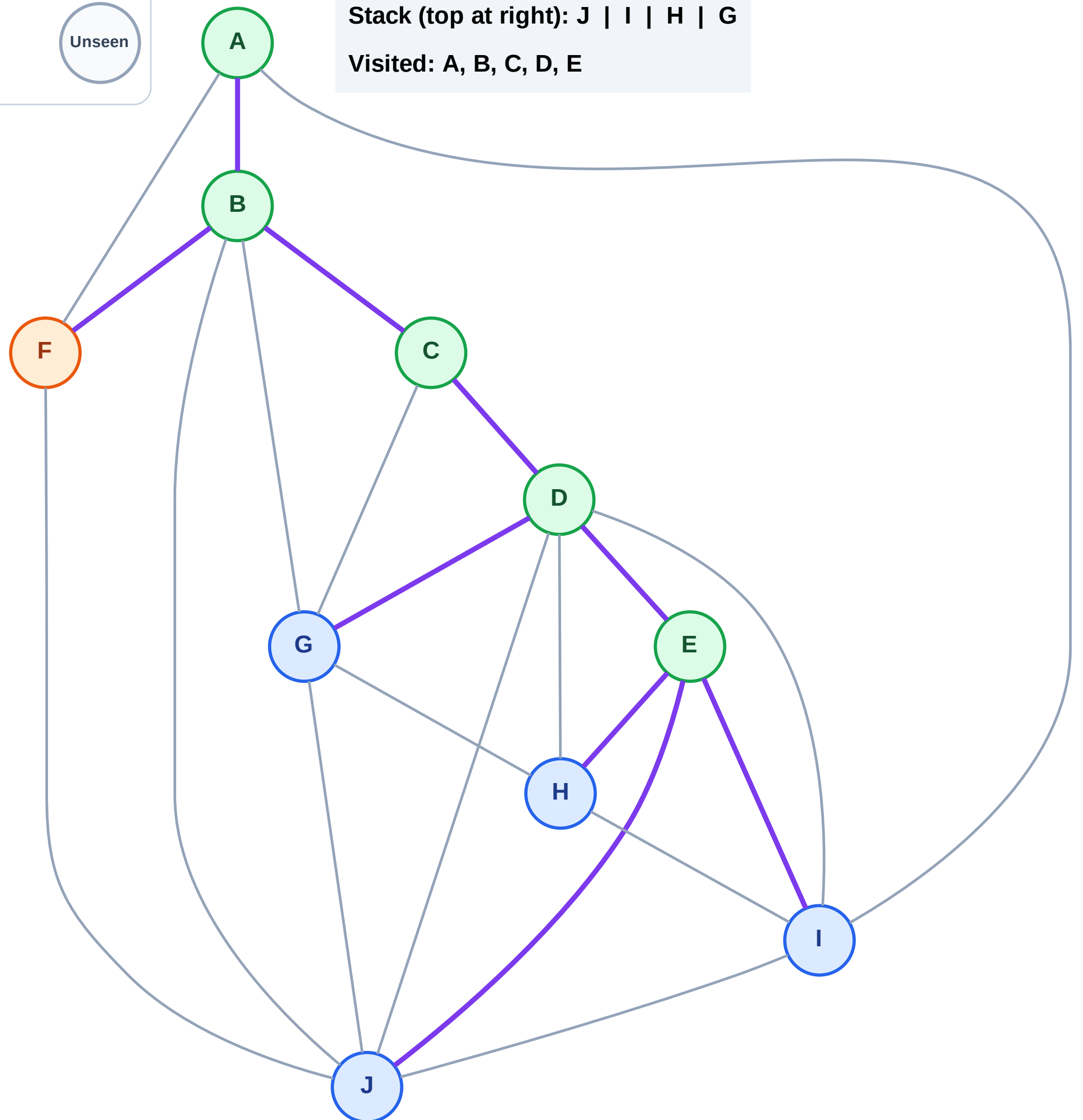
Step 12: Process node F

Legend



Stack (top at right): J | I | H | G

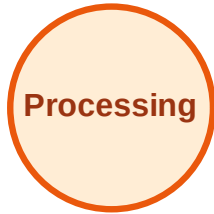
Visited: A, B, C, D, E



DFS Traversal

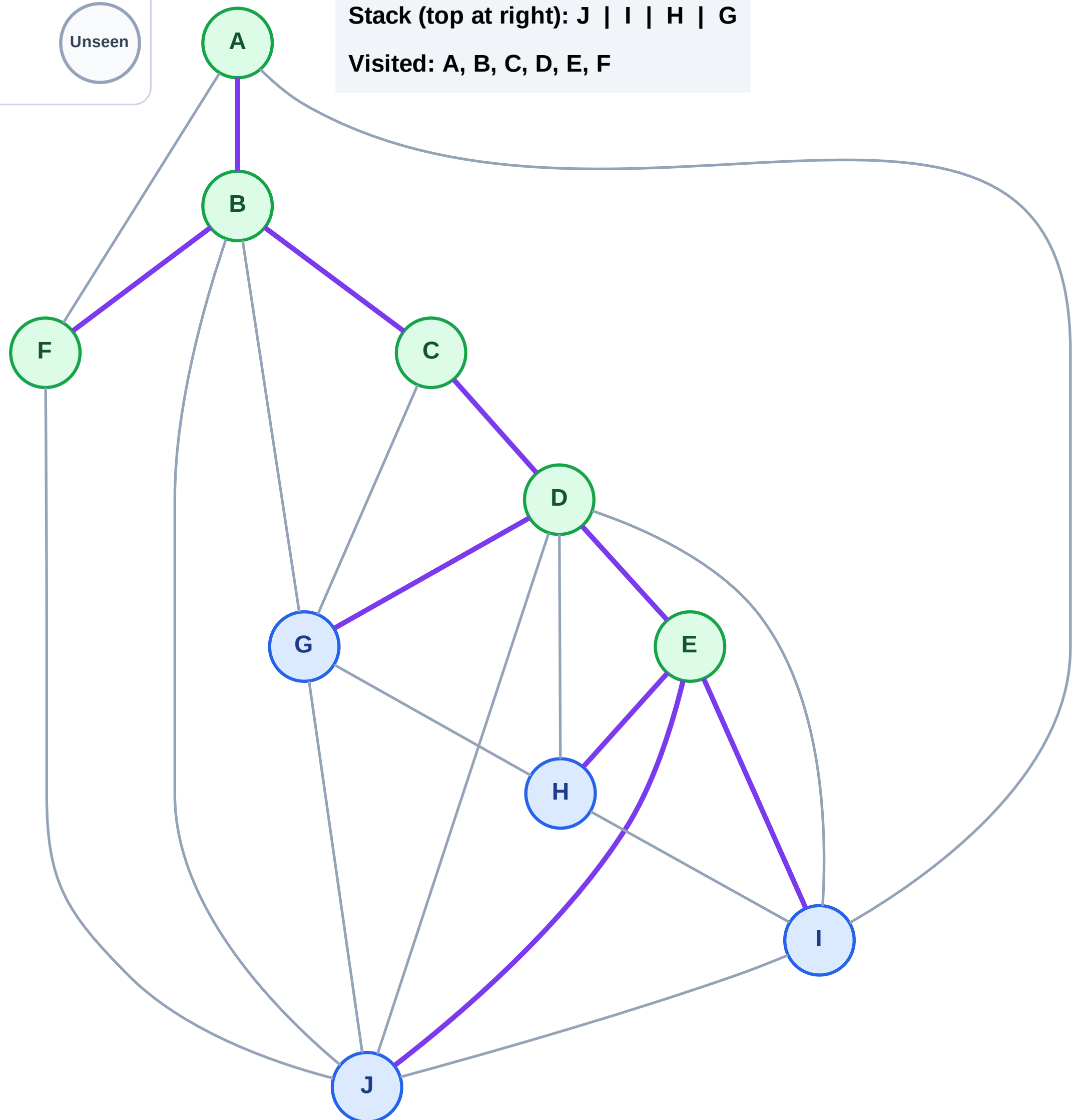
Step 13: No new neighbors from F

Legend



Stack (top at right): J | I | H | G

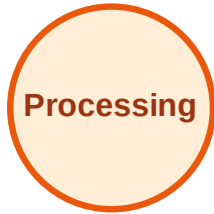
Visited: A, B, C, D, E, F



DFS Traversal

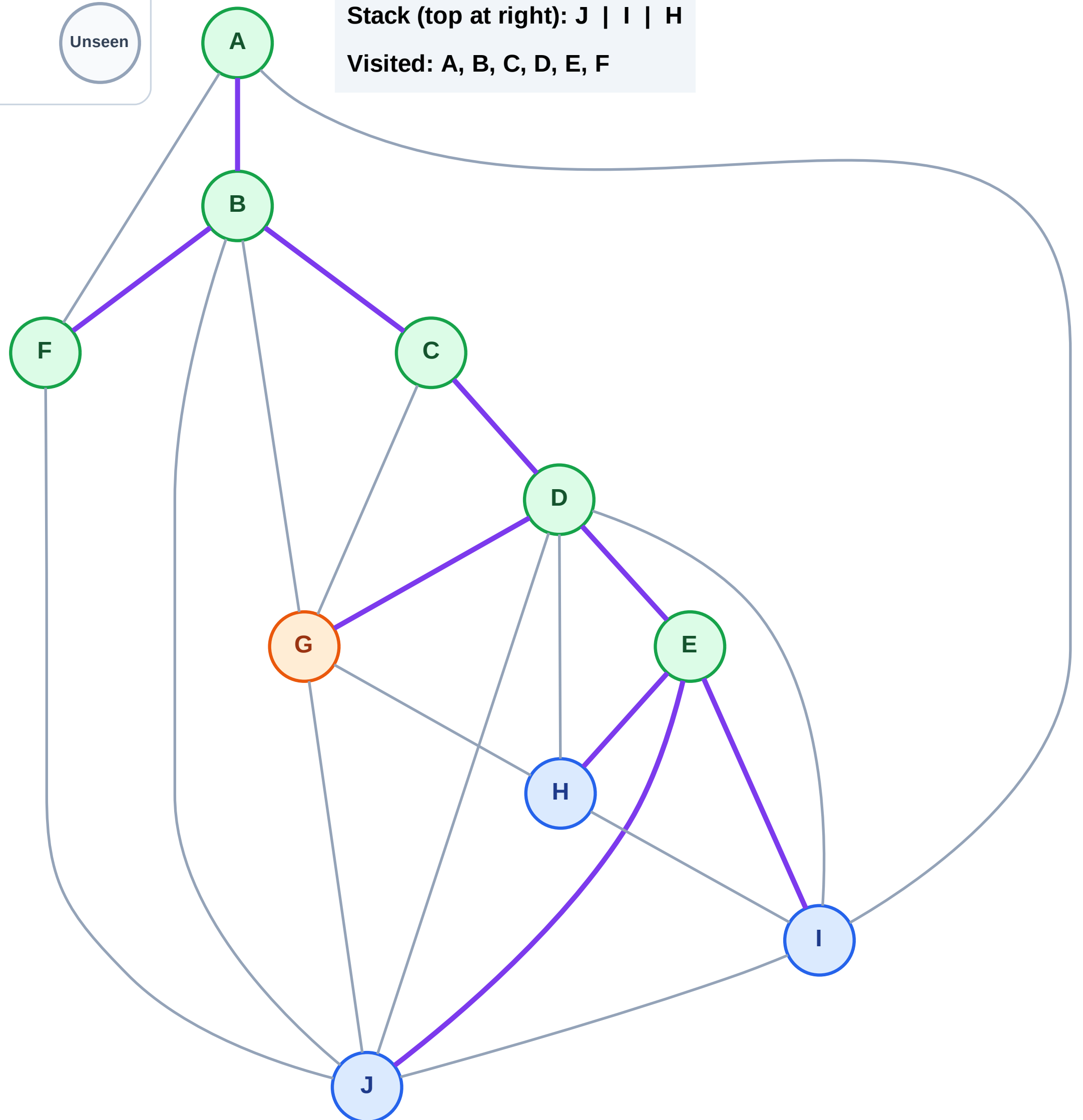
Step 14: Process node G

Legend



Stack (top at right): J | I | H

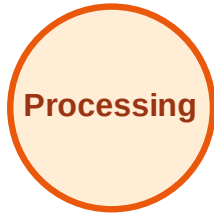
Visited: A, B, C, D, E, F



DFS Traversal

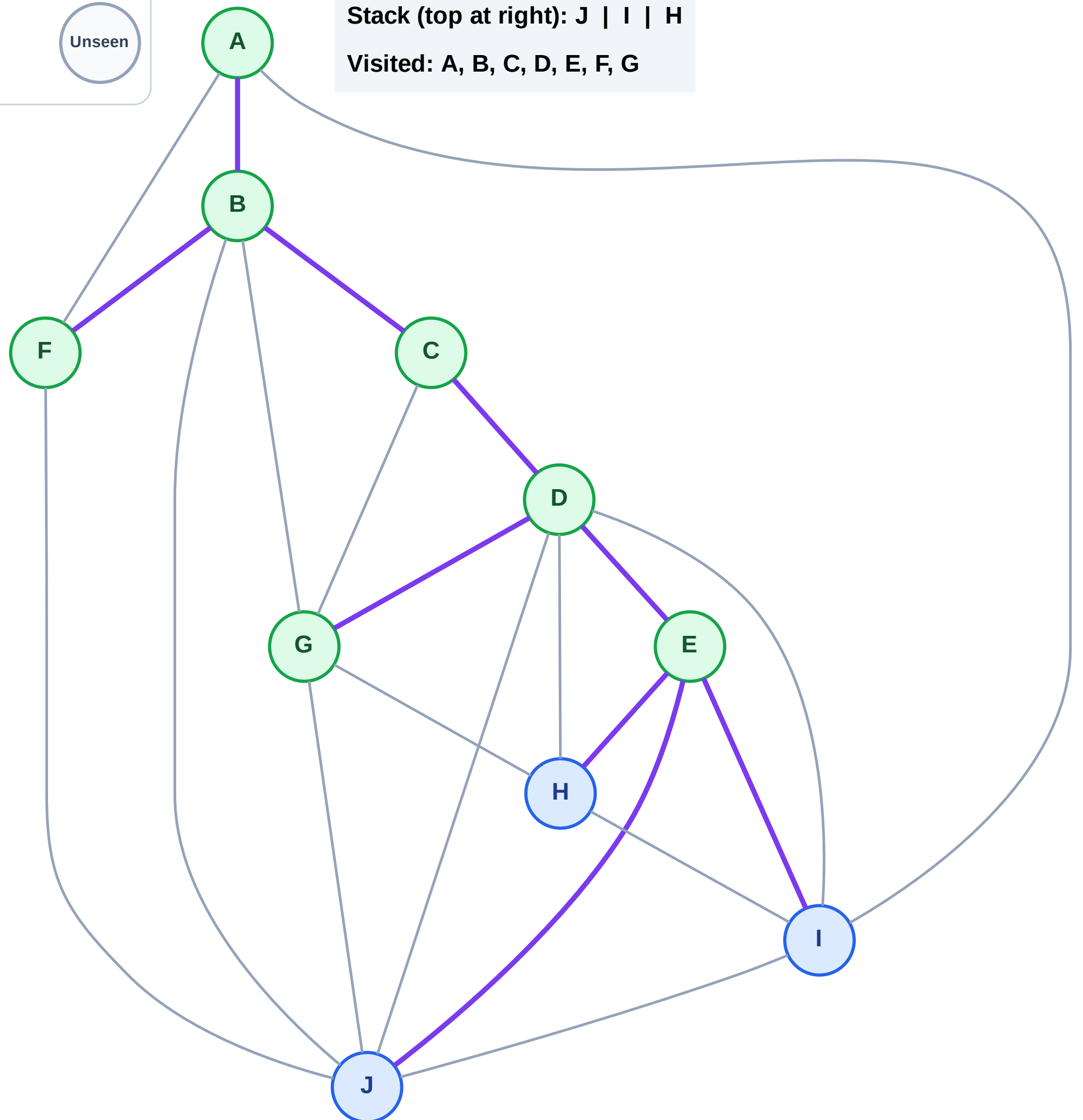
Step 15: No new neighbors from G

Legend



Stack (top at right): J | I | H

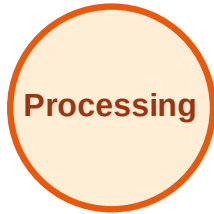
Visited: A, B, C, D, E, F, G



DFS Traversal

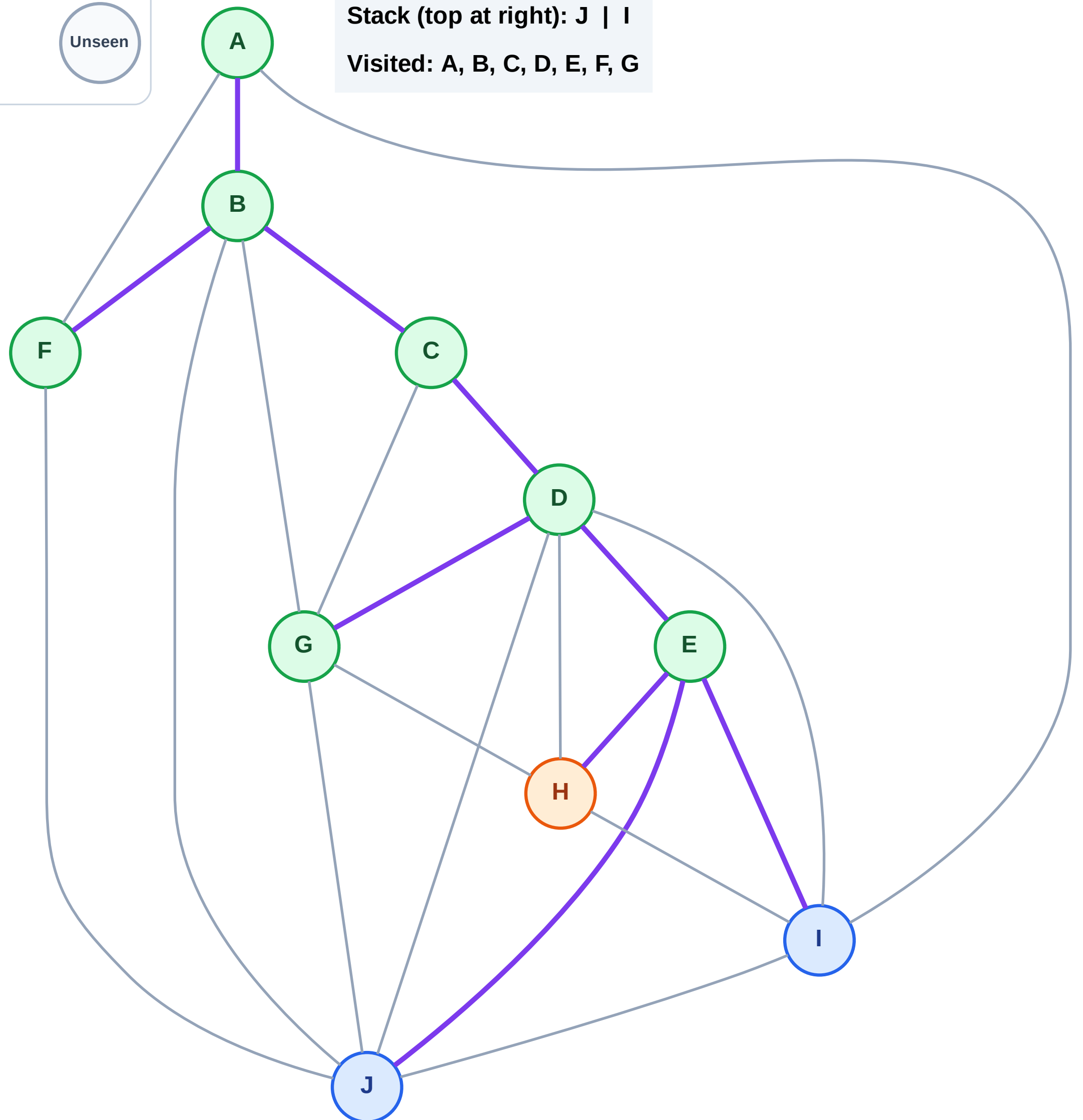
Step 16: Process node H

Legend



Stack (top at right): J | I

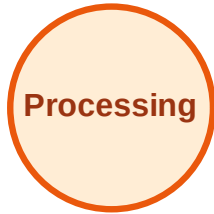
Visited: A, B, C, D, E, F, G



DFS Traversal

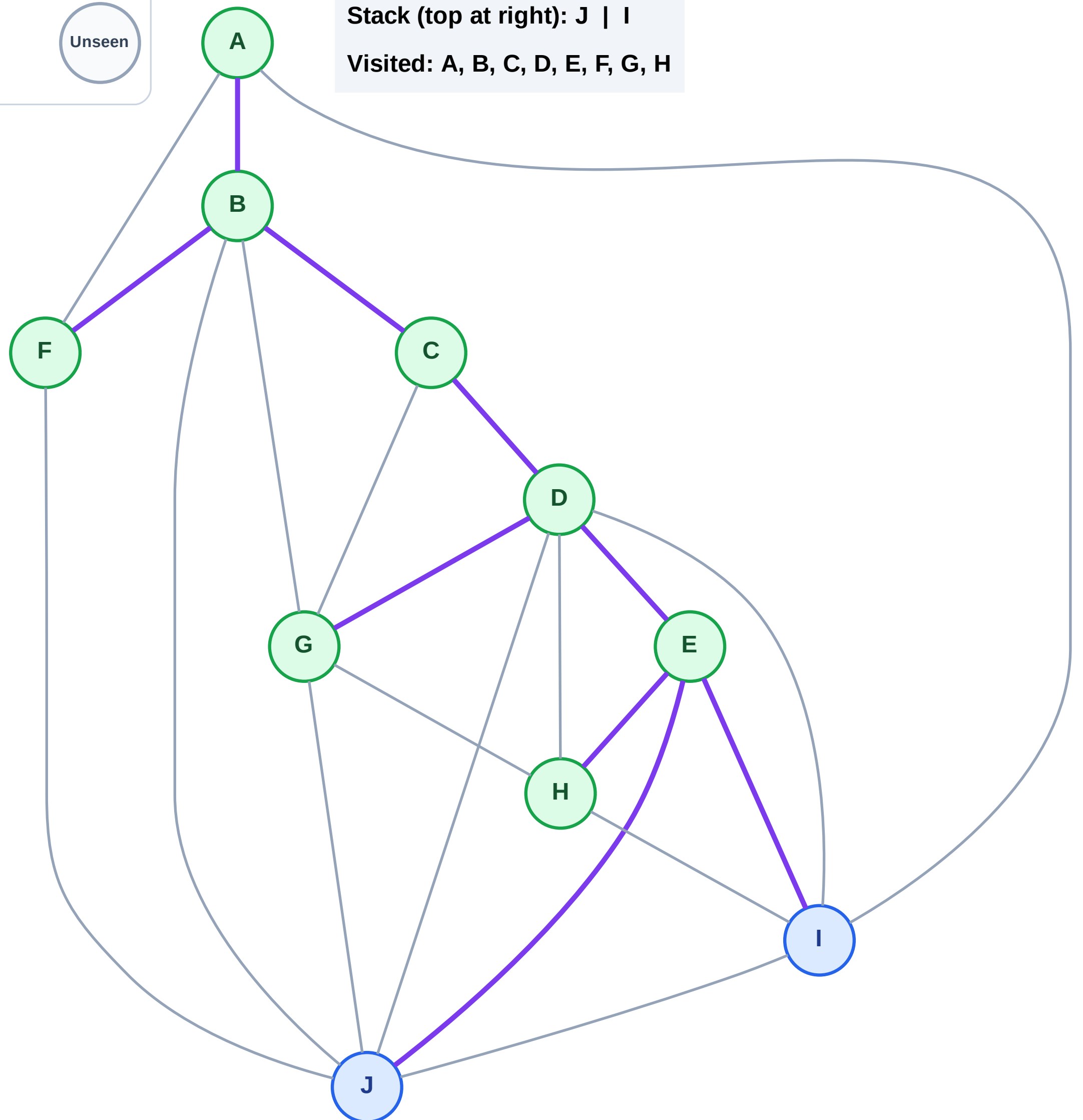
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

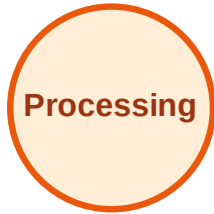
Visited: A, B, C, D, E, F, G, H



DFS Traversal

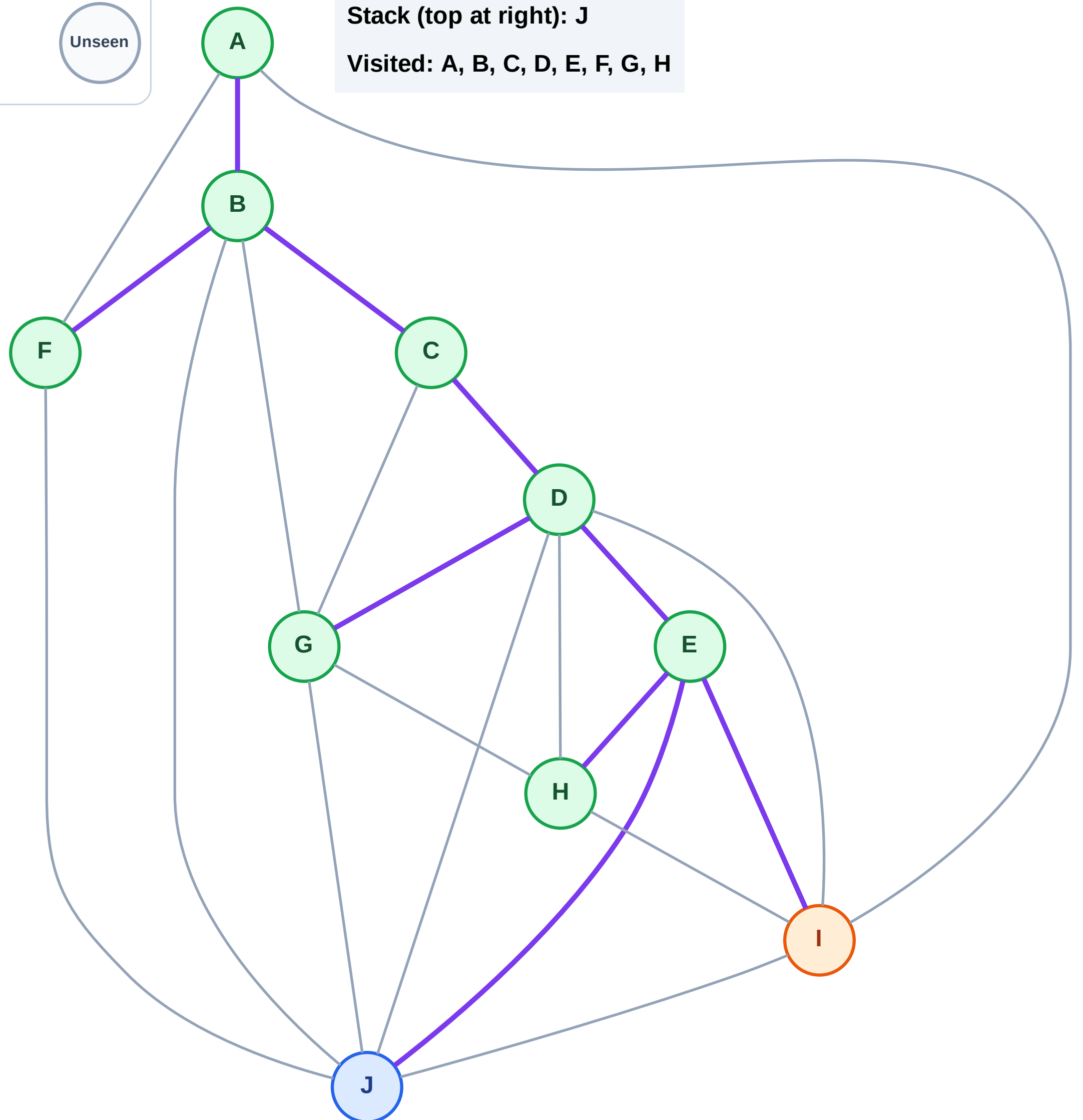
Step 18: Process node I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

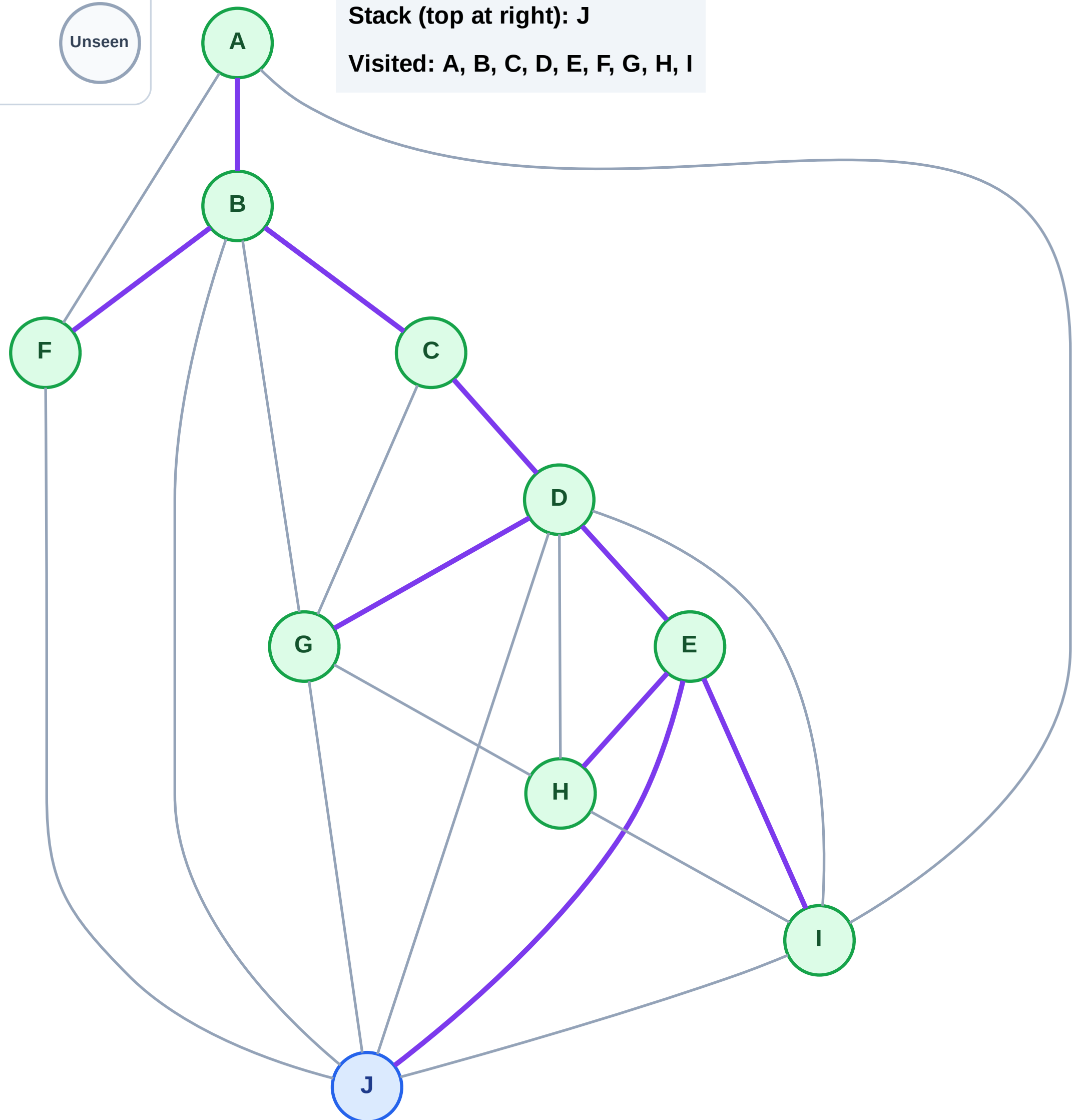
Step 19: No new neighbors from I

Legend



Stack (top at right): J

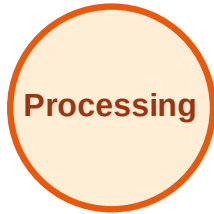
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

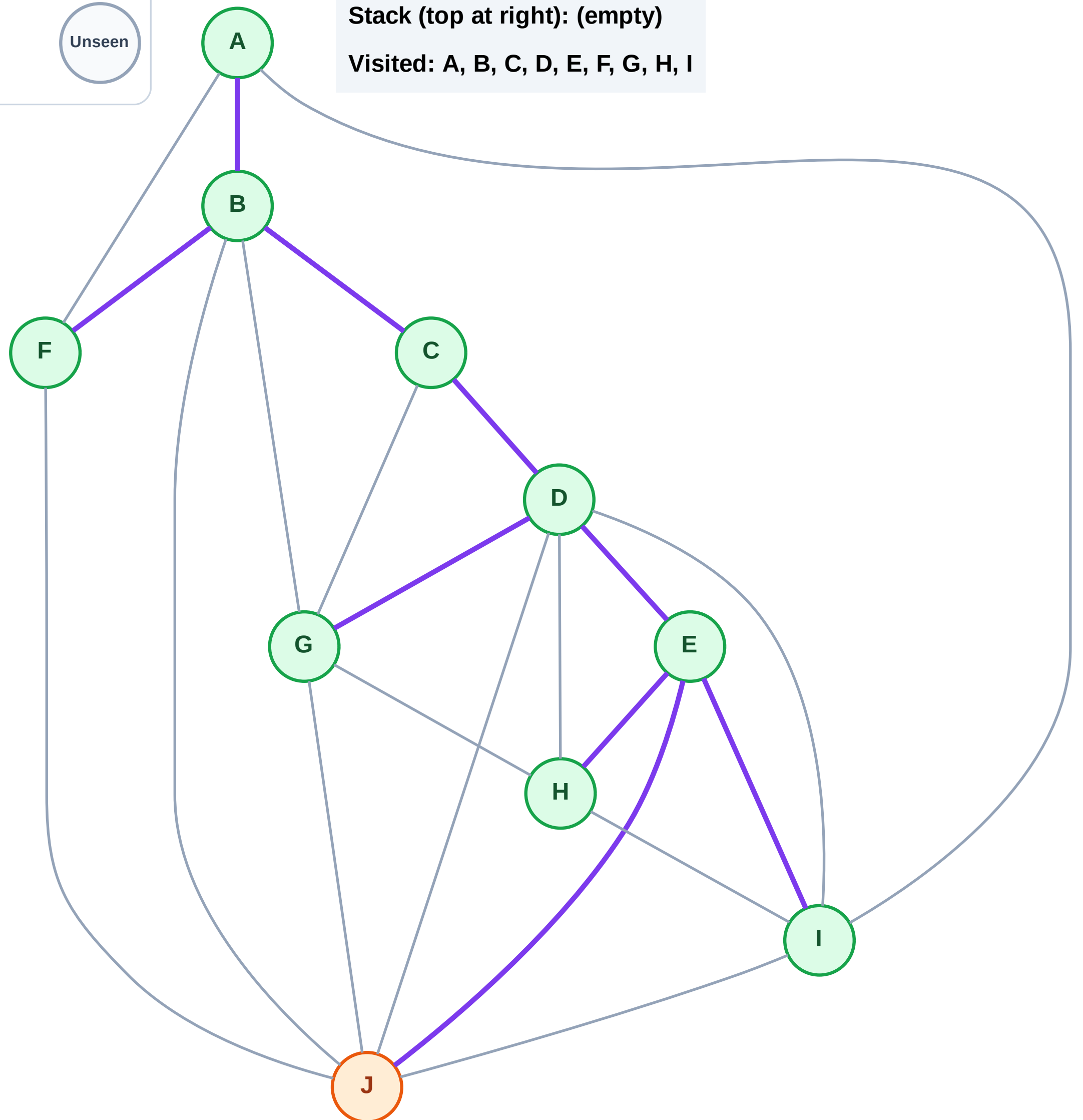
Step 20: Process node J

Legend



Stack (top at right): (empty)

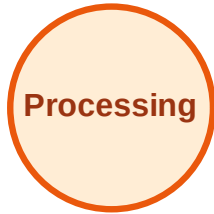
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

